QPL 2026 program (17 – 21 August 2026 in Amsterdam, NL)

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Overview of the program

Welcome to Amsterdam! Registration opens at 8:50 AM on Monday, August 17, at the University of Amsterdam's Roeterseiland campus. The conference runs through Friday, August 21, 2026, with three parallel sessions each afternoon. On Wednesday, there is a boat tour and conference dinner after the parallel sessions. On Thursday, a career fair follows the parallel sessions.

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Monday, August 17th

8:50 – 9:30	Registration
Long plenary talks	
9:30 – 10:15	Alex Maltesson, Ludvig Rodung, Niklas Budinger, Giulia Ferrini, Cameron Calcluth **Equivalence of continuous- and discrete-variable gate-based quantum computers with finite energy**
10:15 – 11:00	Raphaël Mothe, Jessica Bavaresco **Efficient quantum-circuit simulation of classical control of causal order**
Coffee break	
Short plenary talks	
11:30 – 11:55	Vinicius Pretti Rossi, Beata Zjawin, Roberto D. Baldijão, David Schmid, John H. Selby, Ana Belén Sainz **How typical is contextuality?**
11:55 – 12:20	Dichuan Gao, Razin A. Shaikh, Aleks Kissinger **Graphical algebraic geometry: from ideals and varieties to qudit ZH completeness**
Lunch break	

	Parallel A	Parallel B	Parallel C
14:30 – 14:55	Yìlè Yīng, Maria Ciudad Alanon, Daniel Centeno, Jacopo Surace, Marina Maciel Ansanelli, Ruizhi Liu, David Schmid, Robert Spekkens **On whether quantum theory needs complex numbers: the foil theories perspective**	Tomoaki Kawano, Ryo Kashima **Restricted negation in orthomodular logic**	Alexandre Clément **A complete equational theory for real-Clifford+CH quantum circuits**
14:55 – 15:20	Timothée Hoffreumon, Mischa P. Woods **On the experimental falsification of real QT**	Alexandru Baltag, Sonja Smets **Logic meets Wigner's friend (and their friends)**	Xiaoning Bian, Sarah Meng Li, Neil J. Ross, John van de Wetering, Yuming Zhao **A complete and natural rule set for multi-qudit Clifford circuits in all odd prime dimensions**
15:20 – 15:45	Timothée Hoffreumon, Mischa P. Woods **A real matrix theory consistent with QT**	Bert Lindenhovius, Vladimir Zamdzhiev **Operator spaces, linear logic and the Heisenberg–Schrödinger duality of quantum theory**	Colin Blake **Completeness for prime-dimensional phase-affine circuits**
Coffee break			
16:15 – 16:40	Beata Zjawin, Marina Maciel Ansanelli, David Schmid, Yìlè Yīng, John H. Selby, Ciarán M. Gilligan-Lee, Ana Belén Sainz, Robert Spekkens **The resource theory of causal influence and knowledge of causal influence**	Priyaa Varshinee Srinivasan, Jean-Simon Pacaud Lemay, Robin Cockett **Generalized inverses of quantum channels: a categorical perspective**	Fedor Kuyanov, Aleks Kissinger **Efficient classical simulation of low-rank-width quantum circuits using ZX-calculus**

16:40 – 17:05	Leonardo Vaglini, Nasra Daher Ahmed, Ravi Kunjwal **Causal inequalities witness non-stabilizerness without magic**	Andre Kornell, Bert Lindenhovius **The category of quantum graphs is closed**	Kwok Ho Wan, Zhenghao Zhong, Ainhoa Zapirain **Simulating magic state cultivation with few Clifford terms**
17:05 – 17:30	Carla Ferradini, Giulia Mazzola, V. Vilasini **Emergent causal order and time direction: bridging causal models and tensor networks**	James Hefford **Nuclearity and trace in monoidal bicategories with application to extended CFTs**	Mark Koch **Classical Clifford+T sampling without computing marginals**

Tuesday, August 18th

Long plenary talks	
9:30 – 10:15	Maximilian Rüsch, Aleks Kissinger, Benjamin Rodatz / Benjamin Rodatz, Boldizsár Poór, Aleks Kissinger **Completeness for fault equivalence of Clifford ZX diagrams / Fault tolerance by construction**
10:15 – 11:00	Samson Abramsky, Rui Soares Barbosa, Carmen Constantin, Martti Karvonen **Algebraic paradoxes in adaptive quantum computation**
Coffee break	
Short plenary talks	
11:30 – 11:55	Manuel Mekonnen, Thomas D. Galley, Markus P. Müller **Invariance under quantum permutations rules out parastatistics**
11:55 – 12:20	Marek Arsenault, Hlér Kristjánsson **A higher-order perspective on quantum signal processing**
Lunch break	

	Parallel A	Parallel B	Parallel C
14:30 – 14:55	Andrey Boris Khesin, Sarah Meng Li, Boldizsár Poór, Benjamin Rodatz, John van de Wetering, Richie Yeung **SpiderCat: optimal fault-tolerant cat state preparation**	Roberto D. Baldijão, Marco Erba, David Schmid, John Selby, Ana Belén Sainz **Tomographically-nonlocal entanglement**	Matthew Wilson **Agent policies from higher-order causal functions**
14:55 – 15:20	Kwok Ho Wan, Henry Price, Qing Yao **Holographic codes seen through ZX-calculus**	Tim Achenbach, Andreas Bluhm, Leevi Leppäjärvi, Ion Nechita, Martin Plávala **Factorization of multimeters: a unified view on nonclassical quantum phenomena**	V. Vilasini, Lin-Qing Chen, Liuhang Ye, Renato Renner **Events and their localisation are relative to a lab**
15:20 – 15:45	Cole Comfort, Giovanni de Felice **The delayed stabiliser ZX-calculus**	Cihan Okay, Aziz Kharoof **The geometry of fiber products of probability polytopes**	Zixuan Liu, Ognyan Oreshkov **Parity erasure: a foundational principle for indefinite causal order**
Coffee break			
16:15 – 16:40	Vincenzo Fiorentino, Kuntal Sengupta **Superposition and its connections to uncertainty, entanglement and the quantum tensor product**	Tom Williams, Mina Doosti, Farid Shahandeh **Sheaf-theoretic preparation contextuality via stochastic extension**	Luca Apadula, Alexei Grinbaum, Časlav Brukner **Reference frames for process matrices: from coordinate parametrization to spacetime representation**
16:40 – 17:05	Gaurang Agrawal, Matt Wilson **Deriving the generalised Born rule from first principles**	Theodoros Yianni, Farid Shahandeh, Nyan Raess **Linear algebra of generalized contextuality in prepare-transform-measure scenarios**	Raphaël Le Bihan, Alastair Abbott, Mnacho Echenim **Probing the composition of processes with first-order-ISOMIX logic**

17:05 – 17:30	James Hefford, Matt Wilson **Quantum theory can decohere from a causally-indefinite post-quantum theory**	David Schmid, Roberto D. Baldijão, John Selby, Ana Belen Sainz, Robert W. Spekkens **Noncontextuality inequalities for prepare-transform-measure scenarios**	Yassine Benhaj, Kuntal Sengupta, Cyril Branciard **How many systems can be dephased before the quantum switch becomes causally definite?**
Poster session (with reception) 17:30 – 19:30			

Wednesday, August 19th

Long plenary talks	
9:30 – 10:15	Miriam Backens, Simon Perdrix / Miriam Backens **Completeness for flow-preserving rewrite rules / Generating one-way computations with flow: flow-preserving rewriting that ignores the interpretation**
10:15 – 11:00	Quanlong Wang, Richard D. P. East, Razin A. Shaikh, Lia Yeh, Boldizsár Poór, Bob Coecke **Beyond Penrose tensor diagrams with the ZX calculus: applications to quantum computing, quantum machine learning, condensed matter physics, and quantum gravity**
Coffee break	
Industry session 11:30 – 12:20	
Lunch break	

	Parallel A	Parallel B	Parallel C
14:30 – 14:55	Luca Apadula, Alessandro Bisio, Giulio Chiribella, Paolo Perinotti, Kyrylo Simonov **Higher-order transformations of bidirectional quantum processes**	Nathan Claudet, Simon Perdrix **Insights in graph state entanglement via r-local complementation: structure and a quasi-polynomial algorithm**	Pablo Arrighi, Doğukan Bakircioglu, Nathan Houyet **Quantum theory over dual-complex numbers**
14:55 – 15:20	Matthew Wilson, James Hefford, Timothée Hoffreumon **Supermaps on generalised theories**	Piotr Mitosek, Miriam Backens **Working with measurement-based computations on qudits**	Nicolas Moulonguet, Augustin Vanrietvelde **Subsystems as subsets of quantum channels, and the strange case of blind agents**
15:20 – 15:45	Samson Abramsky, Radha Jagadeesan **Essential unitarity for higher-order quantum computation**	Aleks Kissinger, John van de Wetering **ZX-flow: a flexible criterion for deterministic computation with ZX-diagrams**	John Harding, Alexander Wilce **Classical explanations in (and of) general probabilistic theories**
Coffee break			
16:15 – 16:40	Thomas Bartsch, Yuhan Gai, Sakura Schäfer-Nameki **Beyond Wigner – how non-invertible symmetries preserve probabilities**	Haytham McDowall-Rose, Razin A. Shaikh, Lia Yeh **From fermions to qubits: a ZX-calculus perspective**	Amrapali Sen, Flavio Del Santo **Superluminal transformations and indeterminism**
16:40 – 17:05	Vanessa Brzić, Satoshi Yoshida, Mio Murao, Marco Túlio Quintino **Higher-order quantum computing with known input states**	Lia Yeh, Jiaxin Huang, Aleks Kissinger, Sarah Meng Li, John van de Wetering **A three-way normal form for stabiliser codes across ZX diagrams, circuits, and tableaus**	Maarten Grothus, V. Vilasini **Impossibility of superluminal signalling rules out causal loops in conical spacetimes**
Boat tour 17:30 – 18:30			
Conference dinner starting 19:00			

Thursday, August 20th

Long plenary talks	
9:30 – 10:15	Cole Comfort, Robert I. Booth **Denotational semantics for stabiliser quantum programs**
10:15 – 11:00	Kathleen Barsse, Romain Péchoux, Simon Perdrix **Quantum control and general recursion beyond the unitary case**
Coffee break	
Short plenary talk	
11:30 – 11:55	Aabhas Gulati, Ion Nechita, Clément Pellegrini **Entanglement in the Dicke subspace**
Conference photo 11:55 – 12:20	
Lunch break	

	Parallel A	Parallel B	Parallel C
14:30 – 14:55	Arianne Meijer-van de Griend, Leo Becker **Pauli gadget synthesis for gatesets with arbitrary even-arity Clifford gates**	Haruki Emori **Quantum statistical functions**	Nadish de Silva, Santanil Jana, Ming Yin **Three-qubit nonlocality paradoxes: beyond GHZ**
14:55 – 15:20	Soichiro Yamazaki, Seiseki Akibue **Multi-qubit controlled gate with optimal T-count**	Michael Zurel, Jack Davis **Basis-independent stabilizerness and maximally noisy magic states**	Shashaank Khanna, Matthew Pusey, Roger Colbeck **Identifying causal structures which cannot support quantum correlations without fine-tuning**
15:20 – 15:45	Akash Kundu **Tensor and gadget reinforcement learning for improved, hardware-aware quantum architecture search**	Cameron Calcluth, Oliver Hahn, Juani Bermejo Vega, Alessandro Ferraro, Giulia Ferrini **Classical simulation of circuits with realistic odd-dimensional Gottesman–Kitaev–Preskill states**	C. E. Lopetegui-Gonzalez, G. Masse, E. Oudot, U. I. Meyer, F. Centrone, F. Grosshans, P. E. Emeriau, U. Chabaud, M. Walschaers **A unified framework for Bell inequalities from continuous-variable contextuality**
15:45 – 16:10	Mark Deaconu, Nihar Gargava, Amolak Ratan Kalra, Michele Mosca, Jon Yard **Buildings for synthesis with Clifford+R**	Massimo Frigerio, Mattia Walschaers, Andrei Aralov, Carlos Ernesto Lopetegui-Gonzalez, Emilie Gillet **Algebraic techniques for photonic state preparation and characterization**	Martin J. Renner, Edwin Peter Lobo, Arturo Konderak, Remigiusz Augusiak, Antonio Acín **Full nonlocality for non-maximally entangled states**
Coffee break 16:10 – 16:40			
Career fair 16:40 – 19:00			

Friday, August 21st

Long plenary talks	
9:30 – 10:15	Thea Li, Vladimir Zamdzhiev **Quantum coherence spaces revisited: a von Neumann (co)algebraic approach**
10:15 – 11:00	Matilde Baroni, Dominik Leichtle, Ivan Šupić, Damian Markham, Marco Túlio Quintino **Composable simultaneous purification: when all communication scenarios reduce to spatial correlations**
Coffee break	
Business meeting 11:30 – 12:20	
Lunch break	

	Parallel A	Parallel B	Parallel C
14:30 – 14:55	Simon Burton, Hussain Anwar **Meromorphic quantum computing**	Noé Delorme, Simon Perdrix **Diagrammatic reasoning with control as a constructor, applications to quantum circuits**	David Schmid, John H. Selby, Vinicius Pretti Rossi, Roberto D. Baldijão, Ana Belén Sainz **Shadows and subsystems of generalised probabilistic theories: when tomographic incompleteness is not a loophole for contextuality proofs**
14:55 – 15:20	Christine Li, Lia Yeh **Transversal AND in quantum codes**	Chris Heunen, Robin Kaarsgaard, Louis Lemonnier **One rig to control them all**	Jan-Åke Larsson **The contextual Heisenberg microscope**
15:20 – 15:45	Jin Ming Koh, Anqi Gong, Andrei C. Diaconu, Daniel Bochen Tan, Alexandra A. Geim, Michael J. Gullans, Norman Y. Yao, Mikhail D. Lukin, Shayan Majidy **Phantom codes: entangling logical qubits without physical operations**	William Schober, Scott Wesley **A complete equational theory for quantum circuits with generalized control**	Daniel McNulty **Quantifying quantum measurement incompatibility via graph invariants**
Coffee break			
16:15 – 16:40	Vivien Vandaele **Asymptotically optimal quantum circuits for comparators and incrementers**	Sacha Cerf, Harold Ollivier **The perturbative method for quantum correlations**	Raffaele D'Avino, Lorenzo Caramelli, Raja Yehia, Gabriel Senno, Roberto González Pousa, Antonio Acín, Tamás Kriváchy **Device independent quantum key distribution with a single measurement per site**
16:40 – 17:05	Giuseppe De Riso, Giuseppe Catalano, Seth Lloyd, Vittorio Giovannetti, Dario De Santis **A resource-efficient quantum-walker quantum RAM**	Subhendu Bikas Ghosh, Snehasish Roy Chowdhury, Guruprasad Kar, Arup Roy, Tamal Guha, Manik Banik **Strong inequivalence of quantum nonlocal resources**	Paul Becsi, Matthew Joseph Hoban **Bounding classical and quantum correlations in Bayesian networks with quasiprobabilities**
End of QPL 2026. Goodbye!			

Abstracts

Abstracts are reproduced from the front matter of the submitted papers, in the order the talks are scheduled. Where a submission carries no abstract of its own, the abstract of the paper's arXiv version is given instead and labelled with its arXiv id.

Monday, August 17th

Long plenary talks

9:30 – 10:15

Equivalence of continuous- and discrete-variable gate-based quantum computers with finite energy

Alex Maltesson, Ludvig Rodung, Niklas Budinger, Giulia Ferrini, Cameron Calcluth

Continuous systems are studied in many branches of modern physics, such as high-energy physics, cosmology, condensed matter physics, quantum chemistry, and field theories. Such systems are expected to benefit from the substantial advantages in computational power of quantum computers. The continuous-variable paradigm of quantum computation provides the most natural computational formalism for these tasks. However, most existing quantum hardware is based on discrete-variable systems. We address this fundamental discrepancy by providing a rigorous framework for translating native continuous-variable algorithms onto qubit-based quantum processors. This mapping is constructed from a gate-based model of continuous-variable quantum computers, consisting of states and operations built from a polynomial sequence of elementary gates in a finite set, with total energy polynomial in the number of modes. We prove that, under realistic constraints, a gate-based model of continuous-variable quantum computers can be efficiently simulated using discrete-variable devices, thereby establishing a computational equivalence between these paradigms.

10:15 – 11:00

Efficient quantum-circuit simulation of classical control of causal order

Raphaël Mothe, Jessica Bavaresco

We show that quantum circuits with classical control of causal order admit an efficient simulation by fixed-order quantum circuits: any such transformation acting on N independent channels can be deterministically, exactly, and universally simulated using at most N 2 channel queries. For quantum circuits with quantum control of order, we prove a similar simulation result, but restricted to unitary input channels (hence non-universal). We further generalize these constructions to extended, bipartite quantum channels and demonstrate that they can be implemented catalytically, without consuming the additional N 2 −N queries required for the simulation. These results show the limitations of the computational power of quantum circuits with classical control of causal order in general, and of quantum circuits with quantum control of causal order when only their action on unitary channels is considered.

Short plenary talks

11:30 – 11:55

How typical is contextuality?

Vinicius Pretti Rossi, Beata Zjawin, Roberto D. Baldijão, David Schmid, John H. Selby, Ana Belén Sainz

Identifying when observed statistics cannot be explained by any reasonable classical model is a central problem in quantum foundations. A principled and universally applicable approach to defining and identifying nonclassicality is given by the notion of generalised noncontextuality. Here, we study the typicality of contextuality — namely, the likelihood that randomly chosen quantum preparations and measurements produce nonclassical statistics. Using numerical linear programs to test for the existence of a generalised-noncontextual model, we find that contextuality is fairly common: even in experiments with only a modest number of random preparations and measurements, contextuality arises with probability over 99%. We also show that while the typicality of contextuality decreases as the purity (sharpness) of the preparations (measurements) decreases, this dependence is not especially pronounced, so contextuality is fairly typical even in settings with realistic noise. Finally, we show that although nonzero contextuality is quite typical, quantitatively high degrees of contextuality are not as typical, and so large quantum advantages (like for parity-oblivious multiplexing, which we take as a case study) are not as typical. We provide an open-source toolbox that outputs the typicality of contextuality as a function of tunable parameters (such as lower and upper bounds on purity and other constraints on states and measurements). This toolbox can inform the design of experiments that achieve the desired typicality of contextuality for specified experimental constraints.

11:55 – 12:20

Graphical algebraic geometry: from ideals and varieties to qudit ZH completeness

Dichuan Gao, Razin A. Shaikh, Aleks Kissinger

We introduce Graphical Algebraic Geometry (GAG), a family of diagrammatic languages extending the Graphical Linear Algebra programme. We construct several languages within this family and prove that they are universal and complete for the corresponding (co)span semantics of commutative algebras and affine varieties. This framework provides clear graphical representations of algebraic structures -- such as polynomials, ideals, and varieties -- enabling intuitive yet rigorous diagrammatic reasoning. We showcase two practical viewpoints on GAG. First, we show that instances of counting constraint satisfaction problem (#CSP) are recast as rewrite problems of closed diagrams in GAG. This means that deciding rewritability in GAG is #P-hard, and GAG can be viewed as a complete and compositional rewrite system for networks of polynomial constraints. Second, we characterize the qudit ZH calculus, a diagrammatic language for quantum computation, as an extension of Graphical Algebraic Geometry. This establishes the correspondence that Graphical Algebraic Geometry is to the ZH calculus what Graphical Linear Algebra is to the ZX calculus. Using this construction, we show that computing amplitudes in qudit ZH requires only a constant number of queries to a GAG oracle.

Abstract from the arXiv version, arXiv:2605.13993.

Parallel sessions

14:30 – 14:55 · Parallel A

On whether quantum theory needs complex numbers: the foil theories perspective

Yìlè Yīng, Maria Ciudad Alanon, Daniel Centeno, Jacopo Surace, Marina Maciel Ansanelli, Ruizhi Liu, David Schmid, Robert Spekkens

Recent work by Renou et al. (2021) has led to some controversy concerning the question of whether quantum theory requires complex numbers for its formulation. We promote the view that the main result of that work is best understood not as a claim about the relative merits of different representations of quantum theory, but rather as a claim about the possibility of experimentally adjudicating between standard quantum theory and an alternative theory—a foil theory—known as real-amplitude quantum theory (RQT). In particular, the claim is that this adjudication can be achieved given only an assumption about the causal structure of the experiment. Here, we aim to shed some light on why this is possible, by reconceptualizing the comparison of the two theories as an instance of a broader class of such theory comparisons. By recasting RQT as the subtheory of quantum theory that arises by symmetrizing with respect to the collective action of a time-reversal symmetry, we can compare it to other subtheories that arise by symmetrization, but for different symmetries. If the symmetry has a unitary representation, the resulting foil theory is termed a twirled quantum world, and if it does not (as is the case in RQT), the resulting foil theory is termed a swirled quantum world. We show that, in contrast to RQT, there is no possibility of distinguishing any twirled quantum world from quantum theory given only an assumption about causal structure. We also define analogues of twirling and swirling for an arbitrary generalized probabilistic theory and identify certain necessary conditions on a causal structure for it to be able to support a causal compatibility gap between the theory and its symmetrized version. We draw out the implications of these analyses for the question of how a lack of a shared reference frame state features into the possibility of such a gap.

14:30 – 14:55 · Parallel B

Restricted negation in orthomodular logic

Tomoaki Kawano, Ryo Kashima

Orthomodular logic has long been studied as a fundamental logical framework within quantum logic. In addition to its lattice-theoretic foundations, various formulations have been proposed that incorporate specialized implication connectives and more complex logical operators. In this study, we reconstruct orthomodular logic by introducing an operator referred to as restricted negation, which has received comparatively little attention in the existing literature. We demonstrate that this operator provides a natural and expressive basis for formulating the core principles of orthomodular logic. Furthermore, as a main result, we construct new proof system for orthomodular logic founded on this operator.

14:30 – 14:55 · Parallel C

A complete equational theory for real-Clifford+CH quantum circuits

Alexandre Clément

We introduce a complete equational theory for the fragment of quantum circuits generated by the real Clifford gates plus the twoqubit controlled-Hadamard gate. That is, we give a simple set of equalities between circuits of this fragment, and prove that any other true equation can be derived from these. This is the first such completeness result for a finitely-generated, universal fragment of quantum circuits, with no parameterized gates and no need for ancillas.

14:55 – 15:20 · Parallel A

On the experimental falsification of real QT

Timothée Hoffreumon, Mischa P. Woods

Whether the complex numbers of standard quantum theory are experimentally indispensable has remained open for decades. Real quantum theory (RQT), obtained by replacing complex amplitudes with real ones while retaining the usual Kronecker-product composition rule, reproduces all single-party and bipartite Bell correlations of quantum theory (QT), but its lack of local tomography suggested that the two theories might diverge in more general local experiments. This possibility appeared to be confirmed by Renou et al., who argued that a bilocal network experiment can falsify RQT without falsifying QT. Here we show that this conclusion relies on an experimentally untestable assumption. The key distinction is between product-state independence, which constrains the mathematical form of source states, and operational independence, which is defined entirely by the absence of observable cross-source correlations. We prove that, once source independence is imposed operationally, every finite network correlation achievable in QT is also achievable in RQT with the same locality structure of the measurements. We then extend this equivalence to arbitrary finite sequential multipartite protocols involving channels and measurements with prescribed locality structure. Thus, as long as no violation of QT is observed, RQT cannot be experimentally falsified. Our results restore the empirical indistinguishability of QT and RQT, while showing that they support markedly different pictures of the correlation structure underlying the same observed world.

Abstract from the arXiv version, arXiv:2603.19208.

14:55 – 15:20 · Parallel B

Logic meets Wigner's friend (and their friends)

Alexandru Baltag, Sonja Smets

The abstract for this talk at QPL 2026 is based on [6]: https://arxiv.org/abs/2307.01713 In this presentation we look at the interplay of multi-agent epistemic logic and quantum mechanics. We show under which conditions a logically consistent framework can be provided that satisfies both the standard rules and axioms of multi-agent epistemic reasoning and the axiomatic theory that regulates the flow of quantum information. We take a fresh look at Wigner's Friend thought-experiment [31] and some of its more recent variants and extensions [15, 10] , such as the Frauchiger-Renner (FR) Paradox [18]. The questions we answer include: What is the correct epistemic interpretation of the multiplicity of state assignments in these scenarios? Under which conditions can one include classical observers into the quantum state descriptions, in a way that is still compatible with traditional Quantum Mechanics? Under which conditions can one system be admitted as an additional 'observer' from the perspective of another background observer? By answering these questions we provide a principled solution to Wigner Friend-type paradoxes.

14:55 – 15:20 · Parallel C

A complete and natural rule set for multi-qudit Clifford circuits in all odd prime dimensions

Xiaoning Bian, Sarah Meng Li, Neil J. Ross, John van de Wetering, Yuming Zhao

We present a complete set of rewrite rules for multi-qudit Clifford circuits, where qudit denotes a d-level quantum system with d an odd prime. Completeness means that any two Clifford circuits representing the same linear map can be transformed into each other using these rules. In total, there are 19 Clifford relations, each involving at most three qudits and admitting an intuitive interpretation. Our approach leverages the isomorphism between the symplectic group $Sp(2n, \mathbb{Z}_d)$ and the quotient of the Clifford group by the Pauli group. We first derive a complete set of symplectic relations for $Sp(2n, \mathbb{Z}_d)$, and then lift them to Clifford relations by incorporating Pauli corrections. To do this, we introduce a symplectic normal form that captures the stabiliser tableau of a Clifford operator and is unique up to Pauli correction. This simplification enables a streamlined derivation of a complete set of 66 relations, which we further compress to 18 symplectic relations. Our computations in $Sp(2n, \mathbb{Z}_d)$ are formalised in the Agda proof assistant, providing a machine-verified proof of correctness.

15:20 – 15:45 · Parallel A

A real matrix theory consistent with QT

Timothée Hoffreumon, Mischa P. Woods

Quantum theory was radically different from the theories of nature which came before it. One key difference was its use of complex numbers. This opened a longstanding debate over whether quantum theory fundamentally requires complex numbers -- or if their use is merely a convenient choice. Until recently, this question was considered open. However, in a 2021 Nature article, a decisive argument was presented asserting that quantum theory needs complex numbers since real-number quantum theory is inconsistent with the postulates of quantum theory. In this work, we show that this conclusion was premature, and in actual fact, a real-number quantum theory is consistent with the postulates of quantum theory. Our theory retains key features such as representation locality (i.e. local physical operations are represented by local changes to the states). A direct consequence of our results is that quantum theory based on real or complex numbers are experimentally indistinguishable.

Abstract from the arXiv version, arXiv:2504.02808.

15:20 – 15:45 · Parallel B

Operator spaces, linear logic and the Heisenberg–Schrödinger duality of quantum theory

Bert Lindenhovius, Vladimir Zamdzhiev

We show that the category OS of operator spaces, with complete contractions as morphisms, is locally countably presentable and a model of Intuitionistic Linear Logic in the sense of Lafont. We then describe a model of Classical Linear Logic, based on OS, whose duality is compatible with the HeisenbergSchrödinger duality of quantum theory. We explore connections between the Haagerup tensor product on OS and BV-logic and we also show that OS provides a good setting for studying pure state and mixed state quantum information, the interaction between the two, and even higher-order quantum maps such as the quantum switch. This work was published in LICS'25 [1]. A preprint with many omitted proofs is available [2] and we also have a detailed companion paper [3] that provides a proof of the local presentability of OS and proofs that establish other important categorical properties of OS.

15:20 – 15:45 · Parallel C

Completeness for prime-dimensional phase-affine circuits

Colin Blake

Equational reasoning about circuits is central in quantum software for validation, optimisation, and verification. For qubits, the CNOT-dihedral fragment supports efficient rewriting via phase polynomials and layered normal forms, yielding a complete and practically effective equational theory. In this work we generalise that CNOT-dihedral picture from qubits to prime-dimensional qudits. We present a compact PROP for reversible affine circuits over a prime field, with a strict symmetric monoidal semantics into the affine group and a Lafont-style affine normal form. We then adjoin finite-angle diagonal phase generators and organise them by polynomial degree, obtaining linear, quadratic (odd prime), and cubic (prime greater than 3) calculi. Using binomial-basis identities we derive uniform transport rules, establish unique phase-affine normal forms, and prove completeness: semantic equality coincides with derivable equality. This yields a prime-dimensional, phase-polynomial-aligned generalisation of the CNOT-dihedral equational theory.

16:15 – 16:40 · Parallel A

The resource theory of causal influence and knowledge of causal influence

Beata Zjawin, Marina Maciel Ansanelli, David Schmid, Yìlè Yīng, John H. Selby, Ciarán M. Gilligan-Lee, Ana Belén Sainz, Robert Spekkens

Understanding and quantifying causal relationships between variables is essential for reasoning about the physical world. In this work, we develop a resource-theoretic framework to do so. Here, we focus on the simplest nontrivial setting— two (classical) variables that are causally ordered, meaning that the first has the potential to influence the second. First, we introduce the resource theory that directly quantifies causal influence of a functional dependence in this setting and show that the problem of deciding convertibility of resources and identifying a complete set of monotones has a straightforward solution. Then, we introduce the resource theory that arises naturally when one has uncertainty about the functional dependence. We describe a linear program for deciding the question of whether one resource (i.e., probability distribution over functions) can be converted to another. Moreover, for the case where the variables are binary, we identify a triple of monotones that are complete. We provide an interpretation of these monotones and show that resourcefulness consists both of knowledge of the functional relations and of the degree of causal connectivity of the function relating the variables. Finally, we discuss how our resource theory connects to standard methods for quantifying cause-effect relations and Shannon theory, as well as its possible quantum generalization.

16:15 – 16:40 · Parallel B

Generalized inverses of quantum channels: a categorical perspective

Priyaa Varshinee Srinivasan, Jean-Simon Pacaud Lemay, Robin Cockett

A quantum channel is defined as being completely positive (CP) and trace preserving (TP). While not every quantum channel is invertible or reversible, every quantum channel admits various kinds of generalized inverses such as the Moore-Penrose inverse and the Drazin inverse. A generalized inverse of a quantum channel may not itself be a quantum channel: it often fails to be CP. However, generalized inverses still play an important role in quantum error mitigation. Here, because it is often desirable for the generalized inverse of a quantum channel to be at least TP, the Drazin inverse, which is TP, is favoured over the Moore-Penrose inverse, which is not in general TP. In this paper, we take a categorical perspective on generalized inverses of quantum channels. This allows us to give a simple proof of the fact that the Drazin inverse of a quantum channel is always TP. It also allows us to show that for unital quantum channels, the Drazin inverse is also unital. We then generalize this result to dagger Drazin inverses, which allows us to show that for unital quantum channels, the Moore-Penrose inverse is both TP and unital as well. This opens the door to new applications of both the Drazin inverse and Moore-Penrose inverse in quantum information theory and, in particular, in quantum error mitigation.

16:15 – 16:40 · Parallel C

Efficient classical simulation of low-rank-width quantum circuits using ZX-calculus

Fedor Kuyanov, Aleks Kissinger

In this paper, we introduce an algorithm for contracting (i.e. numerically evaluating) ZX-diagrams whose computational complexity is governed by rank-width, a structural graph parameter that behaves nicely under ZX rewrite rules. Our method evaluates a graph-like ZX-diagram in ˜O(4R) time, given its rank-decomposition of width R. By constructing suitable rank-decompositions, we provide a classical simulation algorithm for quantum circuits whose runtime is bounded by both ˜O(2n) and ˜O(2t/2), where n is the number of qubits and t is the number of non-Clifford phase gates. Thus, our method is no slower than either the na¨ıve state-vector simulation or the elementary stabiliser decomposition baseline, and in practice can be even faster when the reduced ZX-diagram has low rank-width. We benchmark our simulation algorithm against Quimb, a popular tensor contraction library, and observe substantial reductions in floating-point operations, often by several orders of magnitude, for random and structured non-Clifford circuits and for random ZX-diagrams.

16:40 – 17:05 · Parallel A

Causal inequalities witness non-stabilizerness without magic

Leonardo Vaglini, Nasra Daher Ahmed, Ravi Kunjwal

[No abstract - no abstract in the submitted paper; no arXiv version found.]

16:40 – 17:05 · Parallel B

The category of quantum graphs is closed

Andre Kornell, Bert Lindenhovius

We introduce a category qGph of quantum graphs, whose definition is motivated entirely from noncommutative geometry. For all quantum graphs G and H in qGph, we then construct a quantum graph $[G,H]$ of homomorphisms from G to H, making qGph a closed symmetric monoidal category. We prove that for all finite graphs G and H, the quantum graph $[G,H]$ is nonempty iff the (G,H)-homomorphism game has a winning quantum strategy, directly generalizing the classical case. The finite quantum graphs in qGph are tracial, real, and self-adjoint, and the morphisms between them are CP morphisms that are adjoint to a unital $*$-homomorphism. We prove that Weaver's two notions of a CP morphism coincide in this context. We also include a short proof that every finite reflexive quantum graph is the confusability quantum graph of a quantum channel.

Abstract from the arXiv version, arXiv:2601.09685.

16:40 – 17:05 · Parallel C

Simulating magic state cultivation with few Clifford terms

Kwok Ho Wan, Zhenghao Zhong, Ainhoa Zapirain

Building upon [arXiv:2509.01224], we present a few methods on how to simulate the non-Clifford $d=5$ magic state cultivation circuits [arXiv:2409.17595] with a sum of ≈ 8 Clifford ZX-diagrams on average, at 0.1% noise. Compared to a magic cat state stabiliser decomposition of all 53 non-Clifford spiders ($6{,}377{,}292$ terms required), this is more than 7×10^{5} times reduction in the number of terms. Our stabiliser decomposition has the advantage of representing the final non-Clifford state (in light of circuit errors) as a sum of Clifford ZX-diagrams. This will be useful in simulating the escape stage of magic state cultivation, where one needs to port the resultant state of cultivation into a larger Clifford circuit with many more qubits. Still, it's necessary to only track ≈ 8 Clifford terms. Our result sheds light on the simulability of operationally relevant, high T-count quantum circuits with some internal structure. Finally, we provide numerical results for full non-Clifford stabiliser rank simulation based on $\mathtt{tsim}$ along with optimisations using our cutting decompositions. Nearly 4×10^{6} shots per second can be obtained on a laptop for the smaller $d = 3$ circuits at SD6 circuit level noise $p=0.0005$, making it only $\sim$ 1.1 times slower than its (circuit-unspecific and un-optimised) fully Clifford proxy simulation via $\mathtt{stim}$ using S gates.

Abstract from the arXiv version, arXiv:2509.08658.

17:05 – 17:30 · Parallel A

Emergent causal order and time direction: bridging causal models and tensor networks

Carla Ferradini, Giulia Mazzola, V. Vilasini

Can the direction of time and the causal structure of space-time be inferred from operational principles? Causal models and tensor networks offer complementary perspectives: the former encodes cause-effect relations via directed graphs, with intrinsic ordering; the latter describes multipartite systems on undirected graphs, without presupposing directionality. We construct two-way mappings between these two frameworks, linking direction agnostic correlation functions and operational notions of signalling. This clarifies the operational meaning of causal influence in tensor networks and introduces discrete "spacetime rotations" of causal models which preserve signalling relations. Applying our framework to holographic tensor networks, we use tools from causal inference, like graph-separation, to analyse emergent causal structures. By permitting cyclic and indefinite causal structures, our results enable transfer of techniques across tensor networks and a range of causality frameworks.

17:05 – 17:30 · Parallel B

Nuclearity and trace in monoidal bicategories with application to extended CFTs

James Hefford

We develop the theory of nuclear and trace 2-ideals in monoidal bicategories. We show that such 2ideals are present in a certain bicategory of conformal cobordisms and in the bicategory of measured categories. Viewing the latter as a model of infinite dimensional 2-Hilbert spaces we give a definition of a once extended conformal field theory as a nuclear 2-functor.

17:05 – 17:30 · Parallel C

Classical Clifford+T sampling without computing marginals

Mark Koch

We introduce a new exact classical sampling algorithm for Clifford+T circuits, called T-by-T sampling. For a circuit with T-count t, it reduces sampling to evaluating 2t amplitudes of circuits with successively increasing T-counts 1,2,...,t, without the need to compute costly marginal probabilities. Crucially, the number of amplitude evaluations is completely independent of the surrounding Clifford structure. This makes T-by-T sampling well suited for use together with low-rank stabiliser decompositions, which enable efficient amplitude evaluations as long as the T-count is not too high. In that sense, T-by-T sampling is an improvement over the gate-by-gate sampling technique introduced by Bravyi, Gosset, and Liu [6], where the number of amplitudes instead scales with the number of non-monomial operations such as Hadamard gates, which can remain large even in circuits with low stabiliser rank. Concretely, T-by-T sampling offers significant speedups in regimes where T gates are embedded within large Clifford computations. This makes it well suited for applications such as simulating quantum error correction experiments, which typically have lower T complexity but involve extensive Clifford operations and measurements. In particular, we show how T-by-T sampling allows us to take samples from recent end-to-end magic state cultivation circuits with escape up to distance 19 in a couple of seconds.

Tuesday, August 18th

Long plenary talks

9:30 – 10:15

Completeness for fault equivalence of Clifford ZX diagrams

Maximilian Rüsch, Aleks Kissinger, Benjamin Rodatz / Benjamin Rodatz, Boldizsár Poór, Aleks Kissinger

Two circuits are considered to be equivalent under noise if the effect of faults on one circuit is no worse than the effect of faults on the other circuit. We call this relationship fault equivalence. Fault equivalence offers a way to transform circuits while provably preserving their fault-tolerant properties, enabling a framework for fault-tolerant circuit synthesis and optimisation that is correct by construction. The ZX calculus offers a diagrammatic way to represent and reason about quantum circuits and is a useful tool for manipulating circuits while preserving fault equivalence. For this, the usual set of ZX rewrites has to be restricted to not only preserve the underlying linear map represented by the diagram, but also fault equivalence. In this work, we provide a set of ZX rewrites that are sound and complete for fault equivalence of Clifford ZX diagrams. This means that any equivalence that can be derived using the proposed rules is certain to be correct, and any correct equivalence can be derived using only these rules. For this, we utilise diagrammatic constructions called fault gadgets to reason about arbitrary, possibly correlated Pauli faults in ZX diagrams. Fault gadgets allow us to separate the diagram into a fault-free part, which captures the noise-free behaviour of a diagram, and a noisy part that enumerates the effects of all possible faults. Using this, we provide a unique normal form for ZX diagrams under noise and show that any diagram can be brought into this normal form using our proposed rule set.

9:30 – 10:15

Fault tolerance by construction

Maximilian Rüsch, Aleks Kissinger, Benjamin Rodatz / Benjamin Rodatz, Boldizsár Poór, Aleks Kissinger

A key challenge in fault-tolerant quantum computing is synthesising and optimising circuits in a noisy environment, as traditional techniques often fail to account for the effect of noise on circuits. In this work, we propose a framework for designing fault-tolerant quantum circuits that are correct by construction. The framework starts with idealised specifications of fault-tolerant gadgets and refines them using provably sound basic transformations. To reason about manipulating circuits while preserving their error correction properties, we define fault equivalence; two circuits are considered fault-equivalent if all undetectable faults on one circuit have a corresponding fault on the other. This guarantees that the effect of undetectable faults on both circuits is the same. We argue that fault equivalence is a concept that is already implicitly present in the literature. Many problems, such as state preparation and syndrome extraction, can be naturally expressed as finding an implementable circuit that is fault-equivalent to an idealised specification. To utilise fault equivalence in a computationally tractable manner, we adapt the ZX calculus, a diagrammatic language for quantum computing. We restrict its rewrite system to not only preserve the underlying linear map but also fault equivalence, i.e. the circuit's behaviour under noise. Enabled by our framework, we verify, optimise and synthesise new and efficient circuits for syndrome extraction and cat state preparation. We anticipate that fault equivalence can capture and unify different approaches in fault-tolerant quantum computing, paving the way for an end-to-end circuit compilation framework.

10:15 – 11:00

Algebraic paradoxes in adaptive quantum computation

Samson Abramsky, Rui Soares Barbosa, Carmen Constantin, Martti Karvonen

Measurement-based quantum computation (MBQC) is a universal model of quantum computation whose full power requires adaptivity. Contextuality is known to power quantum advantage in MBQC, yet it has resisted algebraic analysis in the adaptive setting. We show that if an adaptive $\mathbb{Z}_2$-linear measurement-based quantum computing protocol deterministically computes a non-affine Boolean function, then the underlying quantum resource satisfies an inconsistent set of linear equations. This witnesses an algebraic form of strong contextuality generalising Mermin's All-versus-Nothing arguments. Such algebraic contextuality can be detected cohomologically, resolving an open question posed by Raussendorf, who had established cohomological witnesses of contextuality for non-adaptive protocols, but left the adaptive case open. We prove this result constructively: we model adaptive measurement protocols as ordinary measurements on a larger scenario of tree-like measurements, and explicitly build the inconsistent equations inductively.

Abstract from the arXiv version, arXiv:2607.26157.

Short plenary talks

11:30 – 11:55

Invariance under quantum permutations rules out parastatistics

Manuel Mekonnen, Thomas D. Galley, Markus P. Müller

Quantum systems invariant under particle exchange are either Bosons or Fermions, even though quantum theory in principle admits more general behavior under permutations. But why do we not observe such paraparticles in nature? The analysis of this question was previously limited primarily to specific quantum field theory models. Here we give two distinct model-independent arguments that rule out parastatistics, i.e. fundamentally indistinguishable quantum systems transforming under higher-dimensional representations of the symmetric group, which draw on quantum information theory and recent research on internal quantum reference frames. First, we introduce a notion of complete invariance: quantum systems should not only preserve their local state under permutations, but also the quantum information they carry about other systems, in analogy to the notion of complete positivity in quantum information theory. Second, we demand that quantum systems are invariant under quantum permutations, i.e. permutations conditioned on values of permutationinvariant observables. For both, we show that the respective principle is fulfilled if and only if the particle is a Boson or Fermion. Our results show how quantum reference frames can shed light on a longstanding problem of quantum physics, they underline the crucial role played by the compositional structure of quantum information, and demonstrate the explanatory power but also subtle limitations of recently proposed quantum covariance principles.

11:55 – 12:20

A higher-order perspective on quantum signal processing

Marek Arsenault, Hlér Kristjánsson

[No abstract - no abstract in the submitted paper; no arXiv version found.]

Parallel sessions

14:30 – 14:55 · Parallel A

SpiderCat: optimal fault-tolerant cat state preparation

Andrey Boris Khesin, Sarah Meng Li, Boldizsár Poór, Benjamin Rodatz, John van de Wetering, Richie Yeung

The ability to fault-tolerantly prepare CAT states, also known as multi-qubit GHZ states, is an important primitive for quantum error correction. It is required for Shor-style syndrome extraction, and can also be used as a subroutine for doing fault-tolerant state preparation of CSS codewords. Existing approaches to fault-tolerant CAT state preparations have been found using computationally expensive heuristics involving SAT solving, reinforcement learning, or exhaustive analysis. In this paper, we constructively find optimal circuits for CAT states in a more scalable way. In particular, we derive formal lower bounds on the number of CNOT gates required for circuits implementing n-qubit CAT states that do not spread errors of weight at most t for $1\leq t \leq 5$. We do this by using fault-equivalent rewrites of ZX-diagrams to reduce it to a problem of characterising certain 3-regular simple graphs. We then provide families of such optimal graphs for infinitely many values of n and $t\leq5$. By encoding the construction of optimal graphs as a constraint satisfaction problem we find explicit constructions for circuits that match this lower bound on CNOT count for all $n\leq50$ and $t \leq 5$ and for nearly all pairs (n,t) with $n\leq 100$ and $t\leq 5$ or $n\leq 50$ and $t\leq 7$, significantly extending the regimes that were achievable by previous methods and improving the resource counts for existing constructions. We additionally show how to trade CNOT count against depth, allowing us to construct constant-depth fault-tolerant implementations using $O(n)$ ancilla and $O(n)$ CNOT gates.

Abstract from the arXiv version, arXiv:2603.05391.

14:30 – 14:55 · Parallel B

Tomographically-nonlocal entanglement

Roberto D. Baldijão, Marco Erba, David Schmid, John Selby, Ana Belén Sainz

Entanglement is a central and subtle feature of quantum theory, whose structure and operational behavior can change dramatically when additional physical constraints, such as symmetries or superselection rules, are imposed. Such constraints can give rise to striking and counter-intuitive phenomena, including local broadcasting of entangled states and failures of entanglement monogamy. These effects naturally arise in tomographically nonlocal theories (like real quantum theory, twirled worlds, or fermionic quantum theory), where composite systems possess holistic degrees of freedom that are inaccessible to local measurements. In this work, we study entanglement in such theories within the framework of generalized probabilistic theories. We show that the failure of tomographic locality leads to two qualitatively distinct forms of entanglement, which we term $\textit{tomographically-local}$ entanglement and $ \textit{tomographically-nonlocal}$ entanglement. We analyze the operational consequences of this distinction, proving that tomographically-nonlocal entanglement is useless for Bell nonlocality, steering, and teleportation, but sufficient for dense coding and perfectly secure data hiding. This framework clarifies the origin of several previously puzzling features of entanglement that arise when tomographic locality fails, as can happen even in quantum theory when one considers fermions or fundamental superselection rules.

Abstract from the arXiv version, arXiv:2602.16280.

14:30 – 14:55 · Parallel C

Agent policies from higher-order causal functions

Matthew Wilson

We establish a correspondence between equivalence classes of agent-state policies for deterministic POMDPs and one-input process functions (the classical-deterministic limit of higher-order quantum operations). We use this correspondence to build a bridge between the agent-environment interaction in artificial intelligence, causal structure in the foundations of physics, and logic in computer science. We construct a *-autonomous category PF of types which supports an interpretation of one-step evaluation of policies, and multi-agent observation constraints, into cuts and monoidal products. In terms of types, we develop the correspondence further by identifying observation-independent decentralised POMDPs as the natural domain for the multi-input process functions used to model indefinite causality. We then prove a strict separation between general multi-input process function and definite-ordered process function performance on such dec-POMDPs, by finding an instance for which policies utilizing an indefinite causal structure can achieve greater finite-horizon rewards than policies which are restricted to a fixed background causal structure.

Abstract from the arXiv version, arXiv:2512.10937.

14:55 – 15:20 · Parallel A

Holographic codes seen through ZX-calculus

Kwok Ho Wan, Henry Price, Qing Yao

We re-visit the pentagon holographic quantum error correcting code from a ZX-calculus perspective. By expressing the underlying tensors as ZX-diagrams, we study the stabiliser structure of the code via Pauli webs. In addition, we obtain a diagrammatic understanding of its logical operators, encoding isometries, Rényi entropy and toy models of black holes/wormholes. Then, motivated by the pentagon holographic code's ZX-diagram, we introduce a family of codes constructed from ZX-diagrams on its dual hyperbolic tessellations and study their logical error rates using belief propagation decoders.

Abstract from the arXiv version, arXiv:2601.04467.

14:55 – 15:20 · Parallel B

Factorization of multimeters: a unified view on nonclassical quantum phenomena

Tim Achenbach, Andreas Bluhm, Leevi Leppäjärvi, Ion Nechita, Martin Plávala

Quantum theory exhibits various nonclassical features, such as measurement incompatibility, contextuality, steering, and Bell nonlocality, which distinguish it from classical physics. These phenomena are often studied separately, but they possess deep interconnections. This work introduces a unified mathematical framework based on commuting diagrams that unifies them. By representing collections of measurements (multimeters) as maps to the set of column-stochastic matrices, we show that measurement compatibility and simulability correspond to specific factorizations of these maps through intermediate systems. We apply this framework to put forward connections between different nonclassical notions and provide factorization-based characterizations for steering assemblages and Bell correlations, including a new perspective on the CHSH inequality witnessing measurement incompatibility. We also investigate the symmetric n-extensions of multimeters and no-signaling behaviors and connect these extensions to a notion of n-wise compatibility and to the existence of n-wise LHV models, respectively. Furthermore, we investigate robustness to noise of nonlocal features by examining factorization conditions for maps involving noisy state spaces, providing geometric criteria for when noisy multimeters can be simulated by simpler measurement settings.

Abstract from the arXiv version, arXiv:2504.19865.

14:55 – 15:20 · Parallel C

Events and their localisation are relative to a lab

V. Vilasini, Lin-Qing Chen, Liuhang Ye, Renato Renner

The notions of events and their localisation fundamentally differ between quantum theory and general relativity, reconciling them becomes even more important and challenging in the context of quantum gravity where a classical spacetime background can no longer be assumed. We therefore propose an operational approach drawing from quantum information, to define events and their localisation relative to a Lab, which in particular includes a choice of physical degree of freedom (the reference) providing a generalised notion of "location". We define a property of the reference, relative measurability, that is sensitive to correlations between the Lab's reference and objects of study. Applying this proposal to analyse the quantum switch (QS), a process widely associated with indefinite causal order, we uncover differences between classical and quantum spacetime realisations of QS, rooted in the relative measurability of the associated references and possibilities for agents' interventions. Our analysis also clarifies a longstanding debate on the interpretation of QS experiments, demonstrating how different conclusions stem from distinct assumptions on the Labs. This provides a foundation for a more unified view of events, localisation, and causality across quantum and relativistic domains.

Abstract from the arXiv version, arXiv:2505.21797.

15:20 – 15:45 · Parallel A

The delayed stabiliser ZX-calculus

Cole Comfort, Giovanni de Felice

We introduce the delayed stabiliser ZX-calculus, a formal graphical language for translation-invariant stabiliser quantum processes: enabling compact, finite reasoning about various families of infinite stabiliser-codes, such as quantum convolutional codes and lattice codes. Our calculus extends the odd-prime-dimensional stabiliser ZX-calculus with only a single additional generator, the delay, which feeds data from one time step to the next. First, we give an interpretation of delayed ZX diagrams as sequences of finite approximations obtained by unrolling the diagram in time. We define a notion of observational equivalence between such sequences and prove that, modulo this equivalence relation, each sequence determines a unique infinite-dimensional stabiliser group. Second, we interpret the delay as multiplication by a formal polynomial indeterminate, encoding infinite sequences of Pauli operators as fractions of polynomials. This allows us to represent the infinite stabiliser group of a delayed stabiliser circuit in terms of a finite generating tableau. We connect both semantics via geometric-series expansion, showing that composition in the infinite stabiliser group semantics approximates composition in the generating tableau semantics. Finally, we give an equational theory for delayed stabiliser ZX-calculus, featuring generalised Euler decomposition and colour change rules. We show that this calculus is sound and universal for the generating tableau semantics. Combining delayed ZX calculus with scalable notation [9], we present a novel generalised local complementation rule for translation-invariant graph state. We conjecture that this rule follows from the axioms of our calculus, from which completeness would follow.

15:20 – 15:45 · Parallel B

The geometry of fiber products of probability polytopes

Cihan Okay, Aziz Kharoof

We develop tools for characterizing vertices of fiber products of polytopes and apply them to simplicial distribution polytopes, a class of probability polytopes arising in quantum foundations and quantum information. In the theory of simplicial distributions, a pair of simplicial sets encoding measurement and outcome spaces determines a convex polytope of compatible probability assignments. Our first results give geometric criteria for detecting vertices of fiber products in terms of support data. These results are obtained in the more general framework of inverse limits of diagrams of polytopes in standard form, and they translate to corresponding criteria for simplicial distributions on arbitrary colimits of measurement spaces. We then focus on one-dimensional measurement spaces, where simplicial distributions recover and generalize local marginal polytopes in graphical models. In this setting, our sharpest results concern dipole graphs, for which we obtain a complete characterization of vertices and refine it to a graph-theoretic criterion. These characterizations are reminiscent of the classical support-graph criteria for transportation polytopes, but they arise in a richer class of polytopes in which vertex structure depends not only on support acyclicity but also on additional geometric compatibility data. Using the collapsing method from simplicial topology, we transfer the dipole characterization to rose graphs and obtain analogous results there. Finally, we apply collapsing to complete bipartite graphs, which encode physically relevant bipartite Bell scenarios, and more generally to arbitrary connected graphs, yielding lower bounds on the number of vertices.

Abstract from the arXiv version, arXiv:2603.19479.

15:20 – 15:45 · Parallel C

Parity erasure: a foundational principle for indefinite causal order

Zixuan Liu, Ognyan Oreshkov

Processes with indefinite causal order can arise when quantum theory is locally valid and they allow accomplishing new informational tasks. Despite recent progress, the correlations allowed in such processes have not been clearly understood. Here, we propose to study the constraints on information exchange through such processes in a paradigm of locally sequential operations. In this paradigm, we identify an information-theoretic principle constraining the correlations, termed parity erasure, which follows from the local validity of causality. We show that this principle completely characterizes the local-tomography representation of higher-order processes with indefinite causal order: among all multipartite channels, the ones describing valid higher-order processes are those and only those whose input-output correlations respect parity erasure. This approach reveals a fundamental property of information exchange in scenarios with indefinite causal structure and opens a new arena for exploring their potential applications.

Abstract from the arXiv version, arXiv:2512.08635.

16:15 – 16:40 · Parallel A

Superposition and its connections to uncertainty, entanglement and the quantum tensor product

Vincenzo Fiorentino, Kuntal Sengupta

[No abstract - no abstract in the submitted paper; no arXiv version found.]

16:15 – 16:40 · Parallel B

Sheaf-theoretic preparation contextuality via stochastic extension

Tom Williams, Mina Doosti, Farid Shahandeh

We introduce a preparation-dual notion of contextuality, formulated as an obstruction to stochastic extension. In parallel with the sheaf-theoretic formulation of measurement contextuality, preparation contextuality arises when locally specified preparation statistics cannot be extended to a single global response matrix compatible with all source contexts. Whereas measurement contextuality concerns the incompatibility of restriction maps (marginalisation), the preparation setting requires stochastic extension of partial conditioning data, which is inherently non-unique. We identify minimal structural and preparation compatibility conditions on admissible extension matrices and show that they enforce a rigid product form. This leads to a notion of preparation contextuality in which the absence of any admissible global response representation witnesses contextuality, while preparation compatibility identifies the cases in which this obstruction is nontrivial. The framework is formulated explicitly in matrix form and illustrated by a quantum-mechanical example exhibiting preparation contextuality.

Abstract from the arXiv version, arXiv:2605.00975.

16:15 – 16:40 · Parallel C

Reference frames for process matrices: from coordinate parametrization to spacetime representation

Luca Apadula, Alexei Grinbaum, Časlav Brukner

We study how to implement and transform frame perspectives for quantum processes in the process-matrix formalism. We argue that, for pure processes, the causal reference frames (CRF)and time-delocalized subsystems (TDS) formalisms should be understood as coordinate parametrizations of a single perspective-neutral higher-order object. A genuine perspective arises when one endows the process with additional frame data by choosing an operational foliation into circuit fragments (events). With this distinction, existing no-go results acquire a clear scope: they rule out unitary transformations that preserve time foliation, attempting to switch perspectives while keeping the fragment boundaries -- hence the global past/future partition -- fixed. Focusing on the quantum switch, we construct explicit maps that transform perspectives unitarily at the price of reshuffling the notions of past and future. We then show that unitary transformations between perspectives can also be achieved in a different way, namely by extending the process with subsystems that define quantum reference frames and provide a shared spatiotemporal scaffold. In this extended setting, complementary CRF/TDS perspectives become unitarily related while preserving global past and future. We discuss how this frame-perspectival approach informs the broader question of empirical realizability of abstract process matrices.

Abstract from the arXiv version, arXiv:2604.02873.

16:40 – 17:05 · Parallel A

Deriving the generalised Born rule from first principles

Gaurang Agrawal, Matt Wilson

A basic postulate of modern compositional approaches to generalised physical theories is the generalised Born rule, in which probabilities are postulated to be computable from the composition of states and effects. In this paper we consider whether this postulate, and the strength of the identification between scalars and probabilities, can be argued from basic principles. To this end, we first consider the most naive possible process- theoretic interpretation of textbook quantum theory, in which physical processes (unitaries) along with states and effects (kets and bras) and a probability function from states and effects satisfying just some basic compatibility axioms are identified. We then show that any process theory equipped with such structure is equivalent to an alternative process theory in which the generalised Born rule holds. We proceed to consider introduction of noise into any such theory, and observe that the result of doing so is a strengthening of the identification between scalars and probabilities; from bare monoid homomorphisms to semiring isomorphisms.

Abstract from the arXiv version, arXiv:2511.21355.

16:40 – 17:05 · Parallel B

Linear algebra of generalized contextuality in prepare-transform-measure scenarios

Theodoros Yianni, Farid Shahandeh, Nyan Raess

Generalized contextuality is a canonical distinguishing property of nonclassical generalized probabilistic theories, in particular quantum mechanics. Methods for certification and characterization of generalized contextuality of a given generalized probabilistic theory are well developed for prepare-measure and single-stage prepare-transform-measure scenarios. In a recent work [arXiv:2512.10000], a bottom-up, statistics-first linear-algebraic framework for contextuality in prepare-measure scenarios was introduced. We extend this approach to operational scenarios with sequential transformations with an arbitrary number of stages. We give a full decision procedure for contextuality of such scenarios within operational theories and analyze its computational complexity. In particular, our decision procedure has a complexity linearly exponential in the minimum generalized probabilistic theory (GPT) dimension, and polynomial in the number of procedures. We demonstrate our framework and approach through multiple examples, including Spekkens' toy theory and the 8-state single-qubit stabilizer theory. In particular, we construct an operational theory in which contextuality manifests itself only in the sequential structure of the transformations. Our findings thus shed new light on the significant role of compositional structures in the phenomenon of generalized contextuality.

Abstract from the arXiv version, arXiv:2607.26139.

16:40 – 17:05 · Parallel C

Probing the composition of processes with first-order-ISOMIX logic

Raphaël Le Bihan, Alastair Abbott, Mnacho Echenim

[No abstract - no abstract in the submitted paper; no arXiv version found.]

17:05 – 17:30 · Parallel A

Quantum theory can decohere from a causally-indefinite post-quantum theory

James Hefford, Matt Wilson

We find a process satisfying the axioms of hyper-decoherence which produces standard quantum theory from the theory of quantum boxes (higher-order quantum theory with the non-signalling tensor product). This hyper-decoherence map evades the no-go theorem of Lee and Selby [8] by relaxing constraints on signalling to the past and the uniqueness of purifications. We discuss some natural opposing conclusions: that the existence of this map might be evidence of a genuine hyper-decoherence process producing causal quantum theory from its causally-indefinite higher-order theory; or that it serves as an indication that the axioms of hyper-decoherence might need careful reconsideration, especially regarding the subtle albeit central role that purity plays.

17:05 – 17:30 · Parallel B

Noncontextuality inequalities for prepare-transform-measure scenarios

David Schmid, Roberto D. Baldijão, John Selby, Ana Belen Sainz, Robert W. Spekkens

We provide the first systematic technique for deriving witnesses of contextuality in prepare-transform-measure scenarios. More specifically, we show how linear quantifier elimination can be used to compute a polytope of correlations consistent with generalized noncontextuality in such scenarios. This polytope is specified as a set of noncontextuality inequalities that are necessary and sufficient conditions for observed data in the scenario to admit of a classical explanation relative to any linear operational identities, if one ignores some constraints from diagram preservation. While including these latter constraints generally leads to tighter inequalities, it seems that nonlinear quantifier elimination would be required to systematically include them. We also provide a linear program which can certify the nonclassicality of a set of numerical data arising in a prepare-transform-measure experiment. We apply our results to get a robust noncontextuality inequality for transformations that can be violated within the stabilizer subtheory. Finally, we give a simple algorithm for computing all the linear operational identities holding among a given set of states, of transformations, or of measurements.

Abstract from the arXiv version, arXiv:2407.09624.

17:05 – 17:30 · Parallel C

How many systems can be dephased before the quantum switch becomes causally definite?

Yassine Benhaj, Kuntal Sengupta, Cyril Branciard

Higher order operations within quantum theory feature processes that allow operations to take place without a well-defined causal order [1, 2]. This feature, often referred to as indefinite causal order or causal nonseparability, has recently received much attention, both in quantum foundations, in the understanding of causation, and in its ability to provide leverage in information processing tasks over scenarios where all operations occur in definite order. A natural enquiry is to investigate how much quantumness, or non-classicality is required for a process to feature indefinite causal order.

Wednesday, August 19th

Long plenary talks

9:30 – 10:15

Completeness for flow-preserving rewrite rules

Miriam Backens, Simon Perdrix / Miriam Backens

In the one-way model of measurement-based quantum computation (MBQC), computation proceeds via measurements on some standard resource state. So-called flow conditions ensure that the overall computation is deterministic in a suitable sense, with Pauli flow being the most general of these. Existing work on rewriting MBQC patterns while preserving the existence of flow has focused on rewrites that reduce the number of qubits. In this work, we show that introducing new Z-measured qubits, connected to any subset of the existing qubits, preserves the existence of Pauli flow. Furthermore, we give a unique canonical form for stabilizer ZX-diagrams inspired by recent work of Hu & Khesin. We prove that any MBQC-like stabilizer ZX-diagram with Pauli flow can be rewritten into this canonical form using only rules which preserve the existence of Pauli flow, and that each of these rules can be reversed while also preserving the existence of Pauli flow. Hence we have complete graphical rewriting for MBQC-like stabilizer ZX-diagrams with Pauli flow.

Abstract from the arXiv version, arXiv:2205.02009.

9:30 – 10:15

Generating one-way computations with flow: flow-preserving rewriting that ignores the interpretation

Miriam Backens, Simon Perdrix / Miriam Backens

The one-way model is a universal model of quantum computation, driven by successive adaptive single-qubit measurements on an entangled resource state. Measurements are non-deterministic, yet if the computation satisfies one of several related families of conditions known as 'flows', the computation can be made deterministic overall by modifying later measurements depending on the outcomes of earlier ones. Flow properties also enable efficient translation from one-way computations to circuits, motivating research into rewriting one-way computations while preserving the existence of flow. Existing approaches to flow-preserving rewriting are used for compilation or optimisation and preserve both the interpretation and the existence of flow. Here, we broaden our perspective to consider flow-preserving rewriting that does not necessarily preserve the interpretation, with applications to creating test instances for software that works with flow, as well as to generating ansätze for quantum machine learning. We show that a family of just three flow-preserving rewrite rules suffices to generate any diagram with flow from a trivial diagram with the desired number of inputs and outputs. This rule set is nearly the same as the complete set of flow- and interpretation-preserving rewrite rules for one-way computations in which all measurements are Pauli; and just a small subset of the flow- and interpretation-preserving rewrite rules for arbitrary measurements.

Abstract from the arXiv version, arXiv:2607.03250.

10:15 – 11:00

Beyond Penrose tensor diagrams with the ZX calculus: applications to quantum computing, quantum machine learning, condensed matter physics, and quantum gravity

Quanlong Wang, Richard D. P. East, Razin A. Shaikh, Lia Yeh, Boldizsár Poór, Bob Coecke

We introduce the Spin-ZX calculus as an elevation of Penrose's diagrams and associated binor calculus to the level of a formal diagrammatic language. The power of doing so is illustrated by the variety of scientific areas we apply it to: permutational quantum computing, quantum machine learning, condensed matter physics, and quantum gravity. Respectively, we analyse permutational computing transition amplitudes, evaluate barren plateaus for SU(2) symmetric ansätze, study properties of AKLT states, and derive the minimum quantised volume in loop quantum gravity. Our starting point is the mixed-dimensional ZX calculus, a purely diagrammatic language that has been proven to be complete for finite-dimensional Hilbert spaces. That is, any equation that can be derived in the Hilbert space formalism, can also be derived in the mixed-dimensional ZX calculus. We embed the Spin-ZX calculus inside the mixed-dimensional ZX calculus, rendering it a quantum information flavoured diagrammatic language for the quantum theory of angular momentum, i.e. SU(2) representation theory. We diagrammatically derive the fundamental spin coupling objects - such as Clebsch-Gordan coefficients, symmetrising mappings between qubits and spin spaces, and spin Hamiltonians - under this embedding. Our results establish the Spin-ZX calculus as a powerful tool for representing and computing with SU(2) systems graphically, offering new insights into foundational relationships and paving the way for new diagrammatic algorithms for theoretical physics.

Abstract from the arXiv version, arXiv:2511.06012.

Parallel sessions

14:30 – 14:55 · Parallel A

Higher-order transformations of bidirectional quantum processes

Luca Apadula, Alessandro Bisio, Giulio Chiribella, Paolo Perinotti, Kyrylo Simonov

Bidirectional devices are devices for which the roles of the input and output ports can be exchanged. Mathematically, these devices are described by bistochastic quantum channels, namely completely positive linear maps that are both trace-preserving and identity-preserving. Recently, it has been shown that bidirectional quantum devices can, in principle, be used in ways that are incompatible with a definite input-output direction, giving rise to a new phenomenon called input-output indefiniteness. Here we characterize the most general forms of input-output indefiniteness, associated with a hierarchy of higher-order transformations built from transformations of bistochastic quantum channels. Some levels of the hierarchy correspond to transformations that combine bistochastic channels in a definite causal order, while generally using each channel in an indefinite input-output direction. For other levels of the hierarchy, the indefiniteness can involve both the local input-output direction of each process and the global causal order among the processes. On the foundational side, the hierarchy of higher-order transformations characterized here can be regarded as the largest set of physical processes compatible with a time-symmetric variant of quantum theory, where the possible state transformations are restricted to bistochastic channels.

Abstract from the arXiv version, arXiv:2602.00856.

14:30 – 14:55 · Parallel B

Insights in graph state entanglement via r-local complementation: structure and a quasi-polynomial algorithm

Nathan Claudet, Simon Perdrix

We describe an algorithm with quasi-polynomial runtime $n^{\log_2(n)+O(1)}$ for deciding local unitary (LU) equivalence of graph states. The algorithm builds on a recent graphical characterisation of LU-equivalence via generalised local complementation. By first transforming the corresponding graphs into a standard form using usual local complementations, LU-equivalence reduces to the existence of a single generalised local complementation that maps one graph to the other. We crucially demonstrate that this reduces to solving a system of quasi-polynomially many linear equations, avoiding an exponential blow-up. As a byproduct, we generalise Bouchet's algorithm for deciding local Clifford (LC) equivalence of graph states by allowing the addition of arbitrary linear constraints. We also improve existing bounds on the size of graph states that are LU- but not LC-equivalent. While the smallest known examples involve 27 qubits, and it is established that no such examples exist for up to 8 qubits, we refine this bound by proving that LU- and LC-equivalence coincide for graph states involving up to 19 qubits.

Abstract from the arXiv version, arXiv:2502.06566.

14:30 – 14:55 · Parallel C

Quantum theory over dual-complex numbers

Pablo Arrighi, Doğukan Bakircioglu, Nathan Houyet

We take quantum theory and replace C by C[ε] where ε2 = 0, i.e. we extend quantum theory to the ring of dual complex numbers. The aim is to develop a common language in which to treat continuous quantum physics and discrete quantum models in a unified manner, including their symmetries. Since quantum theory is linear, introducing ε is enough to model infinitesimals. A first objection to this programme is that C[ε] is not a field, since division by ε is undefined, while quantum mechanics typically relies on division. A second objection concerns whether unitarity still makes sense given ε2 = 0. Hence, the core of this work is dedicated to proving that dual-complex quantum theory remains fully consistent. In particular, norm is preserved at all times, and renormalization never requires dividing by an infinitesimal. An equivalence with conventional quantum theory is demonstrated: the dual-complex extension of a parametrized quantum operation automatically provides a linear treatment of its first-order variations. As a first example application, we provide a unified description of both the Dirac equation in the continuum and the Dirac Quantum Walk in the discrete. We establish the discrete Lorentz covariance of the latter.

14:55 – 15:20 · Parallel A

Supermaps on generalised theories

Matthew Wilson, James Hefford, Timothée Hoffreumon

Categorical supermaps generalise higher-order quantum operations from finite-dimensional quantum theory to arbitrary circuit theories. In this paper, we establish the Yoneda lemma for categorical supermaps, which states that whenever a physical theory has a suitable notion of channel-state duality, then categorical supermaps on that theory can be concretely represented in terms of that duality. This lemma eliminates any guesswork or ambiguity when defining the appropriate notion of supermap for these theories. As a concrete application, we show that the categorical supermaps on Boxworld are in general characterised by a physically well-motivated weaking of the recently proposed NSWSE principle for higher-order processes on boxworld. Furthermore, via the same Yoneda lemma we put forward a stable definition for supermaps in real quantum theory.

Abstract from the arXiv version, arXiv:2602.23865.

14:55 – 15:20 · Parallel B

Working with measurement-based computations on qudits

Piotr Mitosek, Miriam Backens

Measurement-based quantum computing is a universal model of quantum computation in which successive product measurements of an entangled resource state drive the computation. The nondeterministic nature of measurements necessitates adaptivity to ensure an overall deterministic computation. Flow structures characterise cases in which such an adaptive correction procedure is possible. Recently, flow has been defined in a setting where the resource states are prime-dimensional qudit graph states rather than the usual qubit graph states. Yet, this qudit flow definition is more burdensome to work with than analogous definitions for qubits. Here, we give a simpler definition of qudit flow and consider various useful properties of this flow, drawing on results for the qubit case. In particular, we show how to focus qudit flow and argue that focused flow is canonical. We improve the previous algebraic formulation to capture focused flow and use it to obtain an O(?3) flow-finding algorithm (where ?is the number of qudits), matching the best known complexity for qubit flows and improving on the previous O(?4) result for qudits. Furthermore, we explore multiple flow-preserving transformations, thus opening a pathway to using flow for optimisation. These transformations include pivoting, removal and insertion of certain types of vertices, and reversibility of flow. Lastly, we propose an algorithmic approach to generating large qudit computations with flow, for testing or machine learning.

14:55 – 15:20 · Parallel C

Subsystems as subsets of quantum channels, and the strange case of blind agents

Nicolas Moulonguet, Augustin Vanrietvelde

[No abstract - no abstract in the submitted paper; no arXiv version found.]

15:20 – 15:45 · Parallel A

Essential unitarity for higher-order quantum computation

Samson Abramsky, Radha Jagadeesan

We develop a semantic framework for higher-order quantum computation based on a boundarycentric presentation of compact closed categories, building on the Kelly–Laplaza construction as reformulated by Abramsky. Morphisms are represented as polarized boundary linkings composed by execution, and a unit-free monoidal sum provides reversible control and branching. Within this setting, we identify a notion of essential unitarity that extends ordinary unitary evolution from first-order quantum processes to higher-order interfaces. At first order, essential unitarity coincides with standard unitarity; for higher-order interfaces, it characterizes when information is preserved relative to the boundary. We prove that essential unitarity is the unique predicate compatible with dagger-monoidal structure, coherence reindexing, and currying, and agreeing with ordinary unitarity at first order, for all morphisms that coherently mix uniform signature blocks—a condition satisfied throughout the quantum core. We show that this framework is expressively sufficient for higher-order quantum computation. In particular, one-slot, equal-size, purity-preserving supermaps admit coherent realizations entirely within the framework, including a coherent quantum switch.

15:20 – 15:45 · Parallel B

ZX-flow: a flexible criterion for deterministic computation with ZX-diagrams

Aleks Kissinger, John van de Wetering

Flow criteria are used to efficiently extract computations, either in the form of measurement patterns or quantum circuits, from ZX-diagrams. Existing criteria such as causal flow, generalised flow, and Pauli flow, were all originally formulated for graph states, so they require ZX-diagrams to be in a very particular graph-state-like form. This form is easily broken by applying basic ZX rules and makes establishing some desirable properties very complicated. Here, we introduce a new "ZXnative" flow criterion called ZX-flow, formulated using a new type of decoration of a ZX-diagram we call Pauli semiwebs. These are a generalisation of Pauli webs, which have recently been used extensively in reasoning about fault-tolerant computations in the ZX-calculus. We show that ZX-flow is straightforwardly preserved by all Clifford rewrites and furthermore that a ZX-diagram has ZXflow if and only if it is Clifford-equivalent to a graph-like ZX-diagram with Pauli flow. Finally, we show that any diagram with ZX-flow can be readily interpreted either as a deterministic measurementbased computation or as a Clifford isometry followed by a sequence of Pauli exponentials. The latter can then be efficiently extracted to a quantum circuit.

15:20 – 15:45 · Parallel C

Classical explanations in (and of) general probabilistic theories

John Harding, Alexander Wilce

We introduce a notion of the ``explanation" of one (generalized) probabilistic model by another as particular kind of span in the category $\Prob$ of probabilistic models and morphisms. We show that explanations compose under a standard pullback construction (notwithstanding that $\Prob$ does not support arbitrary pullbacks). We then show that every locally-finite probabilistic model has a canonical, sharp classical explanation. The construction is functorial, so every locally-finite probabilistic theory has a canonical, sharp classical (though of course, usually non-local) representation.

Abstract from the arXiv version, arXiv:2603.05627.

16:15 – 16:40 · Parallel A

Beyond Wigner – how non-invertible symmetries preserve probabilities

Thomas Bartsch, Yuhan Gai, Sakura Schäfer-Nameki

In recent years, the traditional notion of symmetry in quantum theory was expanded to so-called generalised or categorical symmetries, which, unlike ordinary group symmetries, may be non-invertible. This appears to be at odds with Wigner's theorem, which requires quantum symmetries to be implemented by (anti)unitary -- and hence invertible -- operators in order to preserve probabilities. We resolve this puzzle for (higher) fusion category symmetries $\mathcal{C}$ by proposing that, instead of acting by unitary operators on a fixed Hilbert space, symmetry defects in $\mathcal{C}$ act as isometries between distinct Hilbert spaces constructed from twisted sectors. As a result, we find that non-invertible symmetries naturally act as trace-preserving quantum channels. Crucially, our construction relies on the symmetry category $\mathcal{C}$ being unitary. We illustrate our proposal through several examples that include Tambara-Yamagami, Fibonacci, and Yang-Lee as well as higher categorical symmetries.

Abstract from the arXiv version, arXiv:2602.07110.

16:15 – 16:40 · Parallel B

From fermions to qubits: a ZX-calculus perspective

Haytham McDowall-Rose, Razin A. Shaikh, Lia Yeh

Mapping fermionic systems to qubits on a quantum computer is often the first step for algorithms in quantum chemistry and condensed matter physics. However, it is difficult to reconcile the many different approaches that have been proposed, such as those based on binary matrices, ternary trees, and stabilizer codes. This challenge is further exacerbated by the many ways to describe them -- transformation of Majorana operators, action on Fock states, encoder circuits, and stabilizers of local encodings -- making it challenging to know when the mappings are equivalent. In this work, we present a graphical framework for fermion-to-qubit mappings that streamlines and unifies various representations through the ZX-calculus. To start, we present the correspondence between linear encodings of the Fock basis and phase-free ZX-diagrams. The commutation rules of scalable ZX-calculus allows us to convert the fermionic operators to Pauli operators under any linear encoding. Next, we give a translation from ternary tree mappings to scalable ZX-diagrams, which not only directly represents the encoder map as a CNOT circuit, but also retains the same structure as the tree. Consequently, we graphically prove that ternary tree transformations are equivalent to linear encodings, a recent result by Chiew et al. The scalable ZX representation moreover enables us to construct an algorithm to directly compute the binary matrix for any ternary tree mapping. Lastly, we present the graphical representation of local fermion-to-qubit encodings. Its encoder ZX-diagram has the same connectivity as the interaction graph of the fermionic Hamiltonian and also allows us to easily identify stabilizers of the encoding.

Abstract from the arXiv version, arXiv:2505.06212.

16:15 – 16:40 · Parallel C

Superluminal transformations and indeterminism

Amrapali Sen, Flavio Del Santo

Quantum theory is widely regarded as fundamentally indeterministic, yet classical frameworks can also exhibit indeterminism once infinite information is abandoned. At the same time, relativity is usually taken to forbid superluminal signalling, although Lorentz symmetry formally admits superluminal transformations (SpTs). Dragan and Ekert have argued that SpTs entail a form of indeterminism analogous to that encountered in quantum theory. Here, we derive a theory-independent no-go theorem from a set of natural assumptions: any framework admitting non-order-preserving SpTs must either abandon finite information, relinquish time-symmetric informational content, deny that the past stores memory, or abandon the notion that time determines a preferred causal ordering. In particular, one possible implication is that any theory accommodating SpTs suggests an ontology with unbounded informational content, akin to deterministic classical theories formulated over the real numbers. Consequently, any ontic indeterminacy associated with superluminal transformations cannot originate from finite information.

Abstract from the arXiv version, arXiv:2601.15263.

16:40 – 17:05 · Parallel A

Higher-order quantum computing with known input states

Vanessa Brzić, Satoshi Yoshida, Mio Murao, Marco Túlio Quintino

In higher-order quantum computing (HOQC), one typically considers the universal transformation of unknown quantum operations, treated as blackboxes. It is also implicitly assumed that the resulting operation must act on arbitrary, and thus unknown, input states. In this work, we explore a variant of this framework in which the operation remains unknown, but the input state is fixed and known. We argue that this assumption is well-motivated in certain practical contexts, such as unitary programming, and show that classical knowledge of the input state can significantly enhance performance. We demonstrate that in the SAR protocol, this knowledge leads to an exponential advantage through a repeat-until-success strategy, highlighting the operational power of known-state higher-order transformations. Moreover, this assumption allows us to distinguish between protocols designed for pure, bipartite, and mixed states, and to identify the class of mixed states for which deterministic and exact implementation becomes possible. Finally, in the numerically tested regimes, general strategies offer no advantage over adaptive or sequential ones, suggesting that indefinite causal order does not help in the known-input setting.

16:40 – 17:05 · Parallel B

A three-way normal form for stabiliser codes across ZX diagrams, circuits, and tableaus

Lia Yeh, Jiaxin Huang, Aleks Kissinger, Sarah Meng Li, John van de Wetering

[No abstract - no abstract in the submitted paper; no arXiv version found.]

16:40 – 17:05 · Parallel C

Impossibility of superluminal signalling rules out causal loops in conical spacetimes

Maarten Grothus, V. Vilasini

Contrary to popular belief, in [1] it was shown that it is theoretically possible to have operationally detectable causal loops without violating the principle of no superluminal signalling (NSS) in (1 + 1)-Minkowski spacetime. No such examples are known in (d + 1)-Minkowski spacetimes for d > 1 spatial dimensions. The formalism underlying this finding is known as the affects framework [2], and is based on causal modelling [3, 4] which provides a-priori independent of spacetime, information-theoretic way to define causality. This allows to precisely distinguish causation and signalling (noting that these are not equivalent due to possibility of causal fine-tuning), and as a consequence, embedding these models in spacetime, it was found that NSS does not imply absence of superluminal causation (this is true in all Minkowski spacetimes, not just d = 1). Whether or not such superluminal causal influences compatible with

NSS can be used to form operationally detectable causal loops also in d > 1 dimensions, has remained a key question. Our previous work [5] developed new techniques relating to (1) information-theoretic methods to infer causation from signalling, in presence of operational redundancies and fine-tuning, and (2) ordertheoretic properties of spacetime geometry, in particular identifying such a property conicality satisfied in d > 1 dimensional Minkowski spacetimes but violated in d = 1. Building on this, the current submission resolves the open question showing that in any conical spacetime, no superluminal signalling does rule out all operationally detectable causal loops. This highlights that the relationships between information-theoretic causality and relativistic principles can generally depend on spacetime geometry.

Thursday, August 20th

Long plenary talks

9:30 – 10:15

Denotational semantics for stabiliser quantum programs

Cole Comfort, Robert I. Booth

The stabiliser fragment of quantum theory is a foundational building block for quantum error correction, and hence for the fault-tolerant compilation of quantum programs. In this article, we develop a sound, universal, and complete denotational semantics for stabiliser operations, including measurement, classically controlled Pauli operators, and affine classical computation; supporting an explicit treatment of quantum error-correcting codes. We interpret stabiliser operations as affine relations over finite fields, yielding a semantics that reflects the algebraic structure underlying stabiliser quantum error correction. Because stabiliser quantum mechanics has a well-behaved algebraic structure, our relational semantics is conceptually transparent and computationally tractable when compared to standard denotational models for general quantum programs. We demonstrate the resulting semantics by describing a small assembly language for stabiliser programs with fully-abstract denotational semantics.

10:15 – 11:00

Quantum control and general recursion beyond the unitary case

Kathleen Barsse, Romain Péchoux, Simon Perdrix

Coherent control, aka quantum control, is a central concept in quantum computing that is attracting increasing attention from both the quantum foundations and quantum software communities. Defining coherent control in the presence of recursion and measurement has long been known to be a major challenge. In particular, no-go results have been established for standard semantical domains like completely positive maps. We address this problem by introducing the first quantum programming language with recursion that allows for the coherent control of arbitrary quantum operations. We equip this language with both an operational and a denotational semantics that we prove to be adequate. To design these semantics, we show that combining coherent control, recursion, and measurement crucially requires describing the evolution of subprograms in the absence of input. To address this, the operational semantics takes into account a default evolution branch, while the denotational semantics uses the concept of coherent quantum operation, based on vacuum extensions. We strengthen the validity of our approach by developing an observational equivalence: two programs are equivalent if their probability of termination is the same in any context. The denotational semantics is shown to be fully abstract with respect to this observational equivalence.

Abstract from the arXiv version, arXiv:2507.10466.

Short plenary talk

11:30 – 11:55

Entanglement in the Dicke subspace

Aabhas Gulati, Ion Nechita, Clément Pellegrini

We provide a complete mathematical theory for the entanglement of mixtures of Dicke states. These quantum states form an important subclass of bosonic states arising in the study of indistinguishable particles. We introduce a tensor-based parametrization where the diagonal entries of these states are encoded as a symmetric tensor, enabling a direct translation between entanglement properties and well-studied convex cones of tensors. Our results bridge multipartite entanglement theory with semialgebraic geometry and the theory of completely positive and copositive tensors. This dictionary maps separability to completely positive tensors, the PPT property to moment tensors, entanglement witnesses to copositive tensors, and decomposable witnesses to sum of squares tensors. Using this framework, we construct explicit PPT entangled states in three or more qutrits, disproving a recent conjecture. We establish that PPT entanglement exists for all multipartite systems with local dimension d >= 3 and n >= 3 parties. We also show that, for mixtures of Dicke states, the PPT condition with respect to the most balanced bipartition implies all other PPT conditions. We further connect bosonic extendibility of mixtures of Dicke states to the duals of known hierarchies for non-negative polynomials, such as the ones by Reznick and Polya. We thus provide semidefinite programming relaxations for separability and entanglement testing in the Dicke subspace.

Abstract from the arXiv version, arXiv:2602.15800.

Parallel sessions

14:30 – 14:55 · Parallel A

Pauli gadget synthesis for gatesets with arbitrary even-arity Clifford gates

Arianne Meijer-van de Griend, Leo Becker

We propose an algorithm that can synthesize circuits to gate sets consisting of an 2n-arity Clifford gate and arbitrary single qubit rotations. To do this, we represent the circuit as a sequence of Pauli gadgets and jointly synthesize them by placing the multi-qubit Clifford in the synthesized circuit and pushing its adjoint through the remaining gadgets such that some qubits are disconnected from some gadgets. The pushed gadgets are collected in a Clifford tableau, which is decomposed afterwards. With the help of our algorithm we run experiments where we synthesize circuits to varying sizes of target multi-qubit gates. These experiments indicate that increases in gate size can lead to smaller output circuits, and that different connectivity constraints have meaningful differences in how higher arity gates effect the circuit sized.

14:30 – 14:55 · Parallel B

Quantum statistical functions

Haruki Emori

Statistical functions such as the moment-generating, characteristic, cumulant-generating, and second characteristic functions are standard tools in classical statistics and probability theory. They provide a systematic means to analyze the statistical properties of a system and find applications in diverse fields. While these functions are ubiquitous in classical theory, a quantum counterpart has remained underdeveloped because of the noncommutativity of operators. The absence of such a framework has obscured the connections between statistical quantities and the nonclassical features of quantum mechanics. Here, we construct a framework for quantum statistical functions that addresses these limitations and unifies the languages of quantum statistics. We show that the functions reproduce standard statistical quantities such as expectation values, variance, and covariance upon differentiation. By extending the framework to include pre- and post-selection, we define conditional functions that generate conditional statistical quantities, including the weak value and the weak variance. We further show that multivariable functions, defined with specific operator orderings, correspond to the Kirkwood--Dirac, Margenau--Hill, and Wigner distributions. By generalizing Bochner's theorem within the theory of compactly supported distributions, we obtain a criterion that separates classical statistics from quantum statistics, linking the failure of positive definiteness of the multivariable function to the emergence of quasiprobability. As an application, we import the classical method of moments and generalized method of moments into quantum estimation, introducing quantum estimators that exploit the proposed functions. Our framework reproduces quantum statistical quantities and incorporates the nonclassical features of quasiprobability, providing a basis for further study of quantum statistics.

Abstract from the arXiv version, arXiv:2602.05821.

14:30 – 14:55 · Parallel C

Three-qubit nonlocality paradoxes: beyond GHZ

Nadish de Silva, Santanil Jana, Ming Yin

Quantum nonlocality paradoxes, such as the GHZ paradox, give inequality-free and logically sharp separations between classical and quantum correlations, and play a central role in information-theoretic tasks admitting unconditional quantum advantage. In this work, we introduce a graph-theoretic and logical framework for analysing nonlocality paradoxes. We give an exhaustive classification of all threequbit nonlocality paradoxes that admit biconditional parity proofs. We show that the resulting structures are substantially richer than previously understood, violating regularity conditions underlying previously known families.

14:55 – 15:20 · Parallel A

Multi-qubit controlled gate with optimal T-count

Soichiro Yamazaki, Seiseki Akibue

Controlled gates are key components in various quantum algorithms. Improving on the prior work of Gosset et al., we show that, for an allowed error ε, 3log2(1/ε)+ o(log(1/ε)) T gates are sufficient to approximate most multi-qubit controlled SU(2)s. We also show that this T-count matches the lower bound among approximating circuits that preserve the block-diagonal form. As an application, general controlled gate synthesis and efficient SU(4) gate synthesis are given.

14:55 – 15:20 · Parallel B

Basis-independent stabilizerness and maximally noisy magic states

Michael Zurel, Jack Davis

Absolutely stabilizer states are those that remain convex mixtures of stabilizer states after conjugation by any unitary. Here we give a characterization of such states for multiple qudits of all prime dimensions by introducing a polytope of their allowed spectra. We illustrate this through the examples of one qubit, two qubits, and one qutrit. In particular, the set of absolutely stabilizer states for a single qubit is a ball inscribed in the stabilizer octahedron, but for higher dimensions the geometry is more complicated. For odd-prime-dimensional qudits, we also give a complete characterization of absolutely Wigner-positive states, i.e., states whose Wigner function remains nonnegative after conjugation by any unitary. In so doing, we show there are absolutely Wigner-positive states that are not absolutely stabilizer, which can be seen as a unitarily-invariant version of bound magic. We then study the radii of the largest balls contained in the sets of absolutely stabilizer states and absolutely Wigner-positive states. These radii respectively tell us the lowest possible purity of nonstabilizer and Wigner-negative states. Conversely, we also find the radius of the smallest ball containing the set of absolutely Wigner-positive states, giving a tight purity-based necessary condition thereof.

Abstract from the arXiv version, arXiv:2602.22336.

14:55 – 15:20 · Parallel C

Identifying causal structures which cannot support quantum correlations without fine-tuning

Shashaank Khanna, Matthew Pusey, Roger Colbeck

[No abstract - no abstract in the submitted paper; no arXiv version found.]

15:20 – 15:45 · Parallel A

Tensor and gadget reinforcement learning for improved, hardware-aware quantum architecture search

Akash Kundu

Variational quantum algorithms hold the promise to address meaningful quantum problems already on noisy intermediate-scale quantum hardware. In spite of the promise, they face the challenge of designing quantum circuits that both solve the target problem and comply with device limitations. Quantum architecture search (QAS) automates the design process of quantum circuits, with reinforcement learning (RL) emerging as a promising approach. Yet, RL-based QAS methods encounter significant scalability issues, as computational and training costs grow rapidly with the number of qubits, circuit depth, and hardware noise. To address these challenges, we introduce $\textit{TensorRL-QAS}$, an improved framework that combines tensor network methods with RL for QAS. By warm-starting the QAS with a matrix product state approximation of the target solution, TensorRL-QAS effectively narrows the search space to physically meaningful circuits and accelerates the convergence to the desired solution. Tested on several quantum chemistry problems of up to 12-qubit, TensorRL-QAS achieves up to a 10-fold reduction in CNOT count and circuit depth compared to baseline methods, while maintaining or surpassing chemical accuracy. It reduces classical optimizer function evaluation by up to 100-fold, accelerates training episodes by up to 98$\%$, and can achieve 50$\%$ success probability for 10-qubit systems, far exceeding the $<$1$\%$ rates of baseline. Robustness and versatility are demonstrated both in the noiseless and noisy scenarios, where we report a simulation of an 8-qubit system. Furthermore, TensorRL-QAS demonstrates effectiveness on systems on 20-qubit quantum systems, positioning it as a state-of-the-art quantum circuit discovery framework for near-term hardware and beyond.

Abstract from the arXiv version, arXiv:2505.09371.

15:20 – 15:45 · Parallel B

Classical simulation of circuits with realistic odd-dimensional Gottesman–Kitaev–Preskill states

Cameron Calcluth, Oliver Hahn, Juani Bermejo Vega, Alessandro Ferraro, Giulia Ferrini

Classically simulating circuits with bosonic codes is challenging due to the prohibitive cost of simulating quantum systems with many, possibly infinite, energy levels. We propose an algorithm to simulate circuits with encoded Gottesman-Kitaev-Preskill (GKP) states, specifically for odd-dimensional encoded qudits. Our approach is tailored to be especially effective in the most challenging but practically relevant regime, where the codeword states exhibit high (but finite) squeezing. Our algorithm leverages the Zak-Gross Wigner function introduced by Davis, Fabre, and Chabaud, which represents infinitely squeezed encoded stabilizer states positively. The run-time of the algorithm scales with the negativity of the Wigner function, allowing for efficient simulation of certain large-scale circuits—namely, input stabilizer GKP states undergoing generalized GKP-encoded Clifford operations followed by modular measurement—with a high degree of squeezing. For stabilizer GKP states exhibiting 12 dB of squeezing, our algorithm can simulate circuits with up to 1000 modes with less than double the number of samples required for a single input mode, which

is in stark contrast to existing simulators. Therefore, this approach holds significant potential for benchmarking early implementations of quantum computing architectures utilizing bosonic codes.

15:20 – 15:45 · Parallel C

A unified framework for Bell inequalities from continuous-variable contextuality

C. E. Lopetegui-Gonzalez, G. Masse, E. Oudot, U. I. Meyer, F. Centrone, F. Grosshans, P. E. Emeriau, U. Chabaud, M. Walschaers

Although the original EPR paradox was formulated in terms of position and momentum, most studies of these phenomena have focused on measurement scenarios with only a discrete number of possible measurement outcomes. Here, we present a framework for studying non-locality that is agnostic to the dimension of the physical systems involved, allowing us to probe purely continuous-variable, discrete-variable, or hybrid non-locality. Our approach allows us to find the optimal Bell inequality for any given measurement scenario and quantifies the amount of non-locality that is present in measurement statistics. This formalism unifies the existing literature on continuous-variable non-locality and allows us to identify new states in which Bell non-locality can be probed through homodyne detection. Notably, we find the first example of continuous-variable non-locality that cannot be mapped to a CHSH Bell inequality. Moreover, we provide several examples of simple hybrid DV-CV entangled states that could lead to near-term violation of Bell inequalities.

Abstract from the arXiv version, arXiv:2601.07686.

15:45 – 16:10 · Parallel A

Buildings for synthesis with Clifford+R

Mark Deaconu, Nihar Gargava, Amolak Ratan Kalra, Michele Mosca, Jon Yard

We study the problem of exact synthesis for the Clifford+R gate set and give the explicit structure of the underlying Bruhat-Tits building for this group. In this process, we also give an alternative proof of the arithmetic nature of this gate set.

15:45 – 16:10 · Parallel B

Algebraic techniques for photonic state preparation and characterization

Massimo Frigerio, Mattia Walschaers, Andrei Aralov, Carlos Ernesto Lopetegui-Gonzalez, Emilie Gillet

[No abstract - no abstract in the submitted paper; no arXiv version found.]

15:45 – 16:10 · Parallel C

Full nonlocality for non-maximally entangled states

Martin J. Renner, Edwin Peter Lobo, Arturo Konderak, Remigiusz Augusiak, Antonio Acín

It is a well-established fact that some quantum correlations are nonlocal, meaning they cannot be described by a local hidden variable model. Certain quantum correlations are so strong that they cannot be reproduced even if one allows only a small fraction of the rounds to admit a local hidden variable description. These correlations are called fully nonlocal and they lead to Bell inequalities in which the quantum bound saturates the non-signaling bound. A famous and well-known example is the Peres-Mermin square in which the underlying state is a four dimensional maximally entangled state. Surprisingly, examples of full nonlocality are so-far only known for maximally entangled states. In this work, we show that in every local Hilbert space dimension $d \geq 3$ there are non-maximally entangled states that are fully nonlocal. In fact, we derive simple criteria to ensure full nonlocality that are only based on the smallest and largest Schmidt coefficient. Furthermore, we show that all pure entangled states can be activated to show full nonlocality in the many-copy scenario.

Friday, August 21st

Long plenary talks

9:30 – 10:15

Quantum coherence spaces revisited: a von Neumann (co)algebraic approach

Thea Li, Vladimir Zamdzhiev

We describe a categorical model of MALL (Multiplicative Additive Linear Logic) inspired by the Heisen-bergSchr¨odinger duality of finite-dimensional quantum theory. Proofs of formulas with positive logical polarity correspond to CPTP (completely positive trace-preserving) maps in our model, i.e. the quantum operations in the Schr¨odinger picture, whereas proofs of formulas with negative logical polarity correspond to CPU (completely positive unital) maps, i.e. the quantum operations in the Heisenberg picture. The mathematical development is based on noncommutative geometry and finite-dimensional von Neumann (co)algebras, which can be defined as special kinds of (co)monoid objects internal to the category of finite-dimensional operator spaces. The full version is accepted to FoSSaCS 2026, see [LZ26] for the extended version with appendices.

10:15 – 11:00

Composable simultaneous purification: when all communication scenarios reduce to spatial correlations

Matilde Baroni, Dominik Leichtle, Ivan Šupić, Damian Markham, Marco Túlio Quintino

Bell non-locality is a powerful framework to distinguish classical, quantum and post-quantum resources, which relies on non-communicating players. Under which restriction can we have the same separations, if we allow for communication? Non-signalling state assemblages, and the fact that they can always be simultaneously purified, turned out to be the key element to restrict the simplest bipartite communication scenario, the prepare-and-measure, to the standard bipartite Bell scenario. Yet, many distinctive features of quantum theory are genuinely multipartite and cannot be reduced to two-party behaviour. In this work we are interested in extending this simultaneous purification inspired result to all multipartite communication schemes. As a first step, we unify and extend the simultaneous purification result from states to instruments and super-instruments, which are composable structures, and open up the possibility to explore more complex communication scenarios. Our main contribution is to establish that arbitrary compositions of non-signalling assemblages cannot escape the standard spatial quantum Bell correlations set. As a consequence, any interactive quantum realization of correlations outside of this set must involve at least one signalling assemblage of quantum operations, even when the resulting correlations are non-signalling.

Abstract from the arXiv version, arXiv:2601.05158.

Parallel sessions

14:30 – 14:55 · Parallel A

Meromorphic quantum computing

Simon Burton, Hussain Anwar

We consider the kinematic axioms of quantum mechanics projectively. Instead of normalized (pure) states up to global phase, states become one-dimensional subspaces of vector spaces. This process of projectivization is functorial and lax monoidal. For qubits it identifies the Bloch sphere with the Riemann sphere. We interpret a fragment of the ZXW-calculus projectively and thereby provide an alternate derivation of the arithmetic GHZ/W-calculus of Coecke et al. We find meromorphic functions that characterize the coherent behaviour of circuits for logical state preparation of quantum codes and magic state distillation.

14:30 – 14:55 · Parallel B

Diagrammatic reasoning with control as a constructor, applications to quantum circuits

Noé Delorme, Simon Perdrix

Control is a fundamental concept in quantum and reversible computational models. It enables the conditional application of a transformation to a system, depending on the state of another system. We introduce a general framework for diagrammatic reasoning featuring control as a constructor. To this end, we provide an elementary axiomatisation of control functors, extending the standard formalism of props to controlled props. As an application, we show that controlled props facilitate diagrammatic reasoning for quantum circuits by introducing a simple and complete set of relations involving at most three qubits, whereas in the standard prop setting any complete axiomatisation necessarily requires relations acting on arbitrarily many qubits.

Abstract from the arXiv version, arXiv:2508.21756.

14:30 – 14:55 · Parallel C

Shadows and subsystems of generalised probabilistic theories: when tomographic incompleteness is not a loophole for contextuality proofs

David Schmid, John H. Selby, Vinicius Pretti Rossi, Roberto D. Baldijão, Ana Belén Sainz

It is commonly believed that failures of tomographic completeness undermine assessments of nonclassicality in noncontextuality experiments. In this work, we study how such failures can indeed lead to mistaken assessments of nonclassicality. We then show that proofs of the failure of noncontextuality are robust to a very broad class of failures of tomographic completeness, including the kinds of failures that are likely to occur in real experiments. We do so by showing that such proofs actually rely on a much weaker assumption that we term relative tomographic completeness: namely, that one's experimental procedures are tomographic for each other. Thus, the failure of noncontextuality can be established even with coarse-grained, effective, emergent, or virtual degrees of freedom. This also implies that the existence of a deeper theory of nature (beyond that being probed in one's experiment) does not in and of itself pose any challenge to proofs of nonclassicality. To prove these results, we first introduce a number of useful new concepts within the framework of generalised probabilistic theories (GPTs). Most notably, we introduce the notion of a GPT subsystem, generalising a range of preexisting notions of subsystems (including those arising from tensor products, direct sums, decoherence processes, virtual encodings, and more). We also introduce the notion of a shadow of a GPT fragment, which captures the information lost when one's states and effects are unknowingly not tomographic for one another.

14:55 – 15:20 · Parallel A

Transversal AND in quantum codes

Christine Li, Lia Yeh

The AND gate is not reversible$\unicode{x2014}$on qubits. However, it is reversible on qutrits, making it a building block for efficient simulation of qubit computation using qutrits. We first observe that there are multiple two-qutrit Clifford+T unitaries that realize the AND gate with T-count 3, and its generalizations to n qubits with T-count $3n-3$. Our main result is the construction of a novel qutrit $\mathopen{[\![} 6,2,2 \mathclose{]\!]}$ quantum error-correcting code with a transversal implementation of the AND gate. The key insight in our approach is that a symmetric T-depth one circuit decomposition$\unicode{x2014}$composed of a CX circuit, T and T dagger gates, followed by the CX circuit in reverse$\unicode{x2014}$of a given unitary can be interpreted as a CSS code. We can increase the code distance by augmenting the code circuit with additional stabilizers while preserving the logical gate. This results in a code with a "built-in" transversal implementation of the original unitary, which can be further concatenated to attain a $\mathopen{[\![} 48,2,4 \mathclose{]\!]}$ code with the same transversal logical gate. Furthermore, we present several protocols for mixed qubit-qutrit codes which we call Qubit Subspace Codes, and for magic state distillation and injection.

Abstract from the arXiv version, arXiv:2603.04548.

14:55 – 15:20 · Parallel B

One rig to control them all

Chris Heunen, Robin Kaarsgaard, Louis Lemonnier

Controlled commands—computations whose execution depends on a separate input—play a central role in reversible Boolean circuits and quantum circuits. However, existing formalisms typically treat control only implicitly, entangled with other aspects of computation. From a semantic perspective, control is most naturally expressed in semisimple rig categories, which—unlike standard circuit models such as props—support both parallel and conditional composition. We present a construction that freely adjoins an explicit syntactic notion of control to a circuit theory specified as a suitable prop, subject to eight universally quantified equations. Our main result is that these equations are sound and complete for the intended semantics of control: the resulting theory satisfies a universal property, identifying it exactly as the circuit subtheory of the free semisimple rig completion. The proof combines coherence for rig categories with a new method based on induction over Gray codes. We illustrate the usefulness of the framework by showing that it simplifies several existing sound and complete axiomatisations of quantum circuits, isolating a small and conceptually clean set of generators and equations. In addition, the same equations yield a sound and complete axiomatisation of the multiply controlled Toffoli gate set, that is universal for reversible Boolean circuits.

14:55 – 15:20 · Parallel C

The contextual Heisenberg microscope

Jan-Åke Larsson

The Heisenberg microscope provides a powerful mental image of the measurement process of quantum mechanics (QM), attempting to explain the uncertainty relation through an uncontrollable back-action from the measurement device. However, Heisenberg's proposed back-action uses features that are not present in the QM description of the world, and according to Bohr not present in the world. Therefore, Bohr argues, the mental image proposed by Heisenberg should be avoided. Later developments by Bell and Kochen-Specker shows that a model that contains the features used for the Heisenberg microscope is in principle possible but must necessarily be nonlocal and contextual. In this paper we will re-examine the measurement process within a restriction of QM known as Stabilizer QM (SQM), that still exhibits for example Greenberger-Horne-Zeilinger nonlocality and Peres-Mermin contextuality. The re-examination will use a recent extension of SQM, the Contextual Ontological Model (COM), where the system state gives a complete description of future measurement outcomes reproducing the quantum predictions, including the mentioned phenomena. We will see that this gives a contextual Heisenberg microscope [1] whose back-action can be completely described within COM, and that the associated randomness originates in the initial state of the pointer system, exactly as in the original description of the Heisenberg microscope. The presence of contextuality, usually seen as prohibiting ontological models, suggests that the contextual Heisenberg microscope picture could perhaps be enabled in general QM.

15:20 – 15:45 · Parallel A

Phantom codes: entangling logical qubits without physical operations

Jin Ming Koh, Anqi Gong, Andrei C. Diaconu, Daniel Bochen Tan, Alexandra A. Geim, Michael J. Gullans, Norman Y. Yao, Mikhail D. Lukin, Shayan Majidy

Fault-tolerant logical entangling gates are essential for scalable quantum computing, but are limited by the error rates and overheads of physical two-qubit gates and measurements. To address this limitation, we introduce phantom codes-quantum error-correcting codes that realize entangling gates between all logical qubits in a code block purely through relabelling of physical qubits during compilation, yielding perfect fidelity with no spatial or temporal overhead. We present a systematic study of such codes. First, we identify phantom codes using complementary numerical and analytical approaches. We exhaustively enumerate all 2.71×10^{10} inequivalent CSS codes up to $n=14$ and identify additional instances up to $n=21$ via SAT-based methods. We then construct higher-distance phantom-code families using quantum Reed-Muller codes and the binarization of qudit codes. Across all identified codes, we characterize other supported fault-tolerant logical Clifford and non-Clifford operations. Second, through end-to-end noisy simulations with state preparation, full QEC cycles, and realistic physical error rates, we demonstrate scalable advantages of phantom codes over the surface code across multiple tasks. We observe a one-to-two order-of-magnitude reduction in logical infidelity at comparable qubit overhead for GHZ-state preparation and Trotterized many-body simulation tasks, given a modest preselection acceptance rate. Our work establishes phantom codes as a viable architectural route to fault-tolerant quantum computation with scalable benefits for workloads with dense local entangling structure, and introduces general tools for systematically exploring the broader landscape of quantum error-correcting codes.

Abstract from the arXiv version, arXiv:2601.20927.

15:20 – 15:45 · Parallel B

A complete equational theory for quantum circuits with generalized control

William Schober, Scott Wesley

Controlled subcircuits are commonplace in quantum algorithms as they represent the quantum analogue of branching statements (if then). We introduce a generalization of controlled subcircuits in the form of a binary operation which sends pairs of unitary quantum circuits to a generalized controlled operation, whose semantics are based on matrix exponentials and logarithms. The family of quantum circuits with generalized controlled operations and circuit exponentiation is denoted HQC for Hierarchical Quantum Circuits. We formalize HQC as a modified prop category using a novel category-theoretic construction, and provide a sound and complete equational theory for diagrammatic reasoning in HQC.

15:20 – 15:45 · Parallel C

Quantifying quantum measurement incompatibility via graph invariants

Daniel McNulty

Measurement incompatibility--the impossibility of jointly measuring certain quantum observables--is a fundamental resource for quantum information processing. We develop a graph-theoretic framework for quantifying this resource for large families of binary measurements, including Pauli observables on multi-qubit systems and k-body Majorana observables on n-mode fermionic systems. To each set of observables we associate an anti-commutativity graph, whose vertices represent observables and whose edges indicate pairs that anti-commute. In this representation, the incompatibility robustness--the minimal amount of classical noise required to render the set jointly measurable--becomes a graph invariant. We derive general bounds on this invariant in terms of the Lovász number, clique number, and fractional chromatic number, and show that the Lovász number yields the correct asymptotic scaling for k-body Majorana observables. For line graphs $L(G)$, which naturally arise in the characterisation of exactly solvable spin models, we obtain spectral bounds on the robustness expressed through the energy and skew-energy of the underlying graph G. These bounds become tight for highly symmetric graphs, leading to closed formulas for several graph families. Finally, we identify structural conditions under which the robustness is determined by a simple function of the graph's maximum degree and the number of vertices and edges, and show that such extremal cases occur only when combinatorial structures such as Hadamard, conference or weighing matrices exist.

Abstract from the arXiv version, arXiv:2511.15954.

16:15 – 16:40 · Parallel A

Asymptotically optimal quantum circuits for comparators and incrementers

Vivien Vandaele

We present quantum circuits for comparison and increment operations that achieve an asymptotically optimal gate count of Θ(n) and depth of Θ(log n) over the Clifford+Toffoli gate set, while using a provably minimal number of qubits. We extend these results to classical–quantum comparators, yielding an improved classical– quantum adder with an optimal qubit count. Given the ubiquity of these operations as algorithmic building blocks, our constructions translate directly into reduced circuit complexity for many quantum algorithms. As a notable example, they can be used to improve a space-efficient circuit for Shor's factoring algorithm, reducing circuit depth from O(n3) to O(n2 log2 n) without increasing either the qubit count or the asymptotic gate complexity. Underpinning these results is a general theorem demonstrating how to trade ancilla qubits for control qubits with low overhead in both depth and gate count, providing a broadly applicable tool for quantum circuit design.

16:15 – 16:40 · Parallel B

The perturbative method for quantum correlations

Sacha Cerf, Harold Ollivier

The characterization of the set of quantum correlations Q—the collection of probability distributions achievable by space-like separated parties sharing a quantum state—remains a foundational challenge in quantum information theory. Since Tsirelson's seminal work [1], the study of nonlocality — in several scenarios beyond the Bell case — has largely relied on convex geometric and algebraic methods [2–4], and hierarchies of semidefinite programs stemming from operator algebras [5–8]. However, despite having been thoroughly studied in the bipartite case [9], the geometric structure of Q in the multipartite setting remains unexplored.

16:15 – 16:40 · Parallel C

Device independent quantum key distribution with a single measurement per site

Raffaele D'Avino, Lorenzo Caramelli, Raja Yehia, Gabriel Senno, Roberto González Pousa, Antonio Acín, Tamás Kriváchy

[No abstract - no abstract in the submitted paper; no arXiv version found.]

16:40 – 17:05 · Parallel A

A resource-efficient quantum-walker quantum RAM

Giuseppe De Riso, Giuseppe Catalano, Seth Lloyd, Vittorio Giovannetti, Dario De Santis

Efficient and coherent data retrieval and storage are essential for harnessing quantum algorithms' speedup. Such a fundamental task is addressed by a quantum Random Access Memory (qRAM). Despite their promising scaling properties, current qRAM proposals demand excessive resources and rely on operations beyond the capabilities of current hardware requirements, rendering their practical realization inefficient. We introduce a novel architecture that significantly reduces resource requirements while preserving optimal complexity scaling for quantum queries. Moreover, unlike previous proposals, our algorithm design leverages a simple, repeated operational block based exclusively on local unitary operations and short-range interactions between a limited number of quantum walkers traveling over a single binary tree. This novel approach not only simplifies experimental requirements by reducing the complexity of necessary operations but also enhances the architecture's scalability by ensuring a resource-efficient, modular design that maintains optimal quantum query performance.

Abstract from the arXiv version, arXiv:2508.02855.

16:40 – 17:05 · Parallel B

Strong inequivalence of quantum nonlocal resources

Subhendu Bikas Ghosh, Snehasish Roy Chowdhury, Guruprasad Kar, Arup Roy, Tamal Guha, Manik Banik

In this work [PRL 132, 250205 (2024), arXiv:2310.16386] we present instances of quantum nonlocal correlations that are incomparable in the strongest sense. Specifically, when starting with an arbitrary many copies of one nonlocal correlation, it becomes impossible to obtain even a single copy of the other correlation, and this incomparability holds in both directions. Such incomparable quantum correlations can be obtained even in the simplest Bell scenario, which involves two parties, each having two dichotomic measurements setups. Notably, there exist an uncountable number of such incomparable correlations. Our result challenges the notion of a 'unique gold coin', often referred to as the 'maximally resourceful state', within the framework of the resource theory of quantum nonlocality. To this end, we provide examples of isotropic quantum correlations that cannot be distilled up-to Tsirelson point,and thus partially answer a long standing open question in nonlocality distillation.

16:40 – 17:05 · Parallel C

Bounding classical and quantum correlations in Bayesian networks with quasiprobabilities

Paul Becsi, Matthew Joseph Hoban

Bell's theorem reveals that quantum theory is in tension with classical causal reasoning and, in particular, the notion of local causality. This is now understood as a particular example of non-classicality in the study of correlations in (Bayesian) networks with both unobserved and observed nodes: the correlations are probability distributions over the observed nodes. There is a great deal of work aiming to understand the bounds on quantum and classical correlations in such networks and one approach is to consider outer approximations to the former. Along these lines, we consider quasiprobabilistic models for Bayesian networks, which can be seen as classical models but the probability distributions involving unobserved nodes are "replaced" with quasiprobabilities that respect normalisation but not positivity. We denote the set of correlations resulting from these models as the quasi set. Such models have a history in the study of Bell-type non-classicality where it has been shown that they can produce all non-signalling correlations. We show a generalisation of this result for a broad class of networks, which motivates a conjecture that the quasi set recovers the so called nested Markov model. Our work utilises a connection to tensor network decompositions, which may be of independent interest.

Abstract from the arXiv version, arXiv:2606.23372.